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Contest (1)

template.cpp

15 lines

```
#include <bits/stdc++.h>
#include <cassert>
using namespace std;

#define rep(i, a, b) for(int i = a; i < (b); ++i)
#define all(x) begin(x), end(x)
#define sz(x) (int)(x).size()
typedef long long ll;
typedef pair<int, int> pii;
typedef vector<int> vi;

int main() {
  cin.tie(0)->sync_with_stdio(0);
  cin.exceptions(cin.failbit);
}
```

.bashrc

3 lines

```
alias c='g++ -Wall -Wconversion -Wfatal-errors -g -std=c
    ++17 \
  -fsanitize=undefined,address'
xmodmap -e 'clear lock' -e 'keycode 66=less greater' #
    caps = <>
```

hash.sh

3 lines

```
# Hashes a file, ignoring all whitespace and comments.
    Use for
# verifying that code was correctly typed.
cpp -dD -P -fpreprocessed | tr -d '[:space:]'| md5sum |
    cut -c-6
```

debug.sh

17 lines

```
set -e
g++ code.cpp -o code
g++ gen.cpp -o gen
g++ brute.cpp -o brute
for((i = 1; ; ++i)); do
    ./gen $i > input_file
    ./code < input_file > myAnswer
    ./brute < input_file > correctAnswer
    diff -Z myAnswer correctAnswer > /dev/null || break
    echo "Passed test: "  $i
done
echo "WA on the following test:"
cat input_file
echo "Your answer is:"
cat myAnswer
echo "Correct answer is:"
cat correctAnswer
```

pragma.h

2 lines

```
#pragma GCC optimize("O3,Ofast,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

troubleshoot.txt

1 lines

```
Set and multiset
```

Mathematics (2)

2.1 Math notes

2.1.1 Derangements

Count the number of permutations of length n with 0 fixed points (i.e. $p(i) \neq i$ for all $i \leq n$)

$$f(n) = \begin{cases} 1, & n = 0, \\ 0, & n = 1, \\ (n-1)\big(f(n-1) + f(n-2)\big), & n \geq 2. \end{cases}$$

An equivalent identity is

$$f(n) = n\,f(n-1) + (-1)^n.$$

2.1.2 Catalan Numbers

- Correctly matched parentheses expressions of length $2n$,
- Rooted binary trees with $n+1$ leaves,
- Dyck paths of length $2n$,
- Ways to triangulate a polygon with $n+2$ sides,
- Different binary search trees that can be constructed with n distinct keys.

$$C_n = \frac{1}{n+1}\binom{2n}{n}, \quad n \geq 0.$$

$$C_n = \binom{2n}{n} - \binom{2n}{n+1}, \quad n \geq 0.$$

$$C_0 = 1, \qquad C_{n+1} = \sum_{i=0}^{n} C_i\, C_{n-i}, \quad n \geq 0.$$

The first few values of the sequence (C_n) are

$$1,\ 1,\ 2,\ 5,\ 14,\ 42,\ 132,\ 429,\ 1430,\ 4862,\ \ldots$$

2.1.3 Binet's Formula

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad \text{for} \quad n \geq 2.$$

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}},$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio and $\psi = \frac{1-\sqrt{5}}{2}$ is its conjugate.

2.1.4 Sums

$$1^4 + 2^4 + 3^4 + \cdots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

2.1.5 Stirling number of the second kind

The Stirling number of the second kind, $S(n,k)$, counts the number of ways to partition a set of n objects into k non-empty, unlabeled subsets. The formula is recursive and is given by:

$S(n,k) = k{\cdot}S(n{-}1,k){+}S(n{-}1,k{-}1), \quad S(0,0) = 1, \quad S(n,0) =$

$$S(n,k) = \frac{1}{k!}\sum_{i=0}^{k}(-1)^{k-i}\binom{k}{i}i^n.$$

2.1.6 Mobius

$$s_{f*g}(n) = \sum_{i=1}^{n} \sum_{d|i} g(d) f(\frac{i}{d})$$

$$s_{f*g}(n) = \sum_{d=1}^{n} \sum_{p=1}^{\lfloor \frac{n}{d} \rfloor} g(d) f(p)$$

$$s_{f*g}(n) = g(1) \sum_{p=1}^{n} f(p) + \sum_{d=2}^{n} \sum_{p=1}^{\lfloor \frac{n}{d} \rfloor} g(d) f(p) = g(1) s_f(n) + \sum_{d=2}^{n} g(d) s_f(\lfloor \frac{n}{d} \rfloor)$$

$$s_f(n) = \frac{s_{f*g}(n) - \sum_{d=2}^{n} s_f(\lfloor \frac{n}{d} \rfloor) g(d)}{g(1)}$$

2.1.7 Combinatorics

$$\sum_{k=0}^{x} \frac{\binom{x}{k}}{\binom{n}{k}} = \frac{n+1}{n-x+1}.$$

2.1.8 Useful Geometry Formulas

Triangle Geometry

Heron's Formula. For a triangle with side lengths a, b, c and semiperimeter

$$s = \frac{a+b+c}{2},$$

the area is

$$A = \sqrt{s(s-a)(s-b)(s-c)}.$$

Area Formulas. For a triangle with side lengths a, b, c, height h_a, circumradius R, and inradius r:

$$A = \frac{1}{2} a h_a, \qquad A = rs, \qquad A = \frac{abc}{4R}, \qquad A = \frac{1}{2} ab \sin \alpha.$$

From these, the circumradius and inradius are:

$$R = \frac{abc}{4A}, \qquad r = \frac{A}{s}.$$

Law of Sines.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Law of Cosines.

$$a^2 = b^2 + c^2 - 2bc \cos A, \quad b^2 = c^2 + a^2 - 2ca \cos B, \quad c^2 = a^2 + b^2 - 2ab \cos C.$$

Analytic Geometry

Let $P(x_0, y_0)$ be a point and a line:

$$\ell : \quad ax + by + c = 0$$

Perpendicular Foot from a Point to a Line.

$$H\left(x_0 - \frac{a(ax_0 + by_0 + c)}{a^2 + b^2}, \quad y_0 - \frac{b(ax_0 + by_0 + c)}{a^2 + b^2} \right).$$

Signed Distance from a Point to a Line.

$$d = \frac{ax_0 + by_0 + c}{\sqrt{a^2 + b^2}}.$$

Reflection of a Point Across a Line.

$$P'\left(x_0 - \frac{2a(ax_0 + by_0 + c)}{a^2 + b^2}, \quad y_0 - \frac{2b(ax_0 + by_0 + c)}{a^2 + b^2} \right).$$

Signed Angle Between Two Vectors. For vectors $\vec{u} = (u_x, u_y)$ and $\vec{v} = (v_x, v_y)$, the signed angle θ from $\vec{u}$ to $\vec{v}$ is

$$\theta = \operatorname{atan2}(u_x v_y - u_y v_x,\ u_x v_x + u_y v_y).$$

Equivalently,

$$\tan \theta = \frac{\vec{u} \times \vec{v}}{\vec{u} \cdot \vec{v}} = \frac{u_x v_y - u_y v_x}{u_x v_x + u_y v_y}.$$

2.2 Algorithms

MillerRabin.h

Description: Primality check for numbers up to $7 \cdot 10^{18}$, extend bases for larger number

Time: $\mathcal{O}(7 \log maxValue)$

Status: Tested on judge.yosupo.jp

fec81f, 36 lines

```
namespace MillerRabin {
  using T = unsigned long long;
  using Int = __uint128_t;

  T binpow(T x, T y, T mod) {
    T ans = 1; x %= mod;
    while (y > 0) {
      if (y & 1) ans = (Int)ans * x % mod;
      x = (Int)x * x % mod, y >>= 1;
    }
    return ans;
  }

  bool notPrime(T n, T base, T d, int s) {
    if (base % n == 0) return false;
    T x = binpow(base % n, d, n);
    if (x == 1 || x == n - 1) return false;
    for (int r = 1; r < s; ++r) {
      x = (Int)x * x % n;
      if (x == n - 1) return false;
    }
    return true;
  }

  bool isPrime(T n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    int s = __builtin_ctzll(n - 1); T d = n >> s;

    const T bases[] = {2, 325, 9375, 28178, 450775,
        9780504, 1795265022};

    for (T base : bases) {
      if (notPrime(n, base, d, s)) return false;
    }
    return true;
  }
}
```

FastFourierTransform.h

Description: FFT to multiply two polynomials.

Usage: Set MOD. Rounded convolution coefficients must fit in a signed 64-bit integer.

Time: $\mathcal{O}(N \log N)$ where N is the total sizes of two polynomials

Status: Well-tested

4551f1, 79 lines

```
namespace FFT {
  using ld = long double;
  const int MOD = 1000000007; // Set to the required
      modulus.
  const ld pi = atan2(1, 1) * 4;

  struct cmplx {
    ld a, b;

    cmplx() {
      a = 0, b = 0;
    }

    cmplx(ld a1, ld b1) {
      a = a1, b = b1;
    }
  };

  inline cmplx sum(cmplx& c1, cmplx& c2) {
    return {c1.a + c2.a, c1.b + c2.b};
```

```
}

inline cmplx sub(cmplx& c1, cmplx& c2) {
  return {c1.a - c2.a, c1.b - c2.b};
}

inline cmplx mult(cmplx& c1, cmplx& c2) {
  return {c1.a * c2.a - c1.b * c2.b, c1.a * c2.b + c1.
      b * c2.a};
}

void fft(vector<cmplx> &a, int n) {
  for (int i = 1, j = 0; i < n; ++i) {
    int bit = n >> 1;
    for (; j >= bit; bit >>= 1) j -= bit;
    j += bit;
    if (i < j) swap(a[i], a[j]);
  }
  for (int k = 1; k < n; k *= 2) {
    ld phi = pi / k;
    cmplx wt(cos(phi), sin(phi));
    for (int i = 0; i < n; i += 2 * k) {
      cmplx w(1, 0);
      for (int j = 0; j < k; ++j) {
        cmplx u = a[i + j], v = mult(a[i + j + k], w);
        a[i + j] = sum(u, v);
        a[i + j + k] = sub(u, v);
        w = mult(w, wt);
      }
    }
  }
}

vector<int> multiply(vector<int> &s, vector<int> &t) {
  if (s.empty() || t.empty()) return {};
  int need = (int) s.size() + t.size() - 1, sz = 1;
  while (sz < need) sz <<= 1;
  vector<cmplx> a(sz), b(sz);
  for (int i = 0; i < sz; ++i) {
    if(i < (int)s.size()) a[i] = cmplx(s[i], 0);
    else a[i] = cmplx(0, 0);
    if(i < (int)t.size()) b[i] = cmplx(t[i], 0);
    else b[i] = cmplx(0, 0);
  }
  fft(a, sz);
  fft(b, sz);
  for (int i = 0; i < sz; ++i) {
    a[i] = mult(a[i], b[i]);
  }

  fft(a, sz);
  reverse(a.begin() + 1, a.end());

  vector<int> ans(need);
  for (int i = 0; i < need; ++i) ans[i] = (llround(a[i
      ].a / sz) % MOD + MOD) % MOD;
    return ans;
  }
}
vector<int> operator * (vector<int> &s, vector<int> &t)
    {
  return FFT::multiply(s, t);
}
```

GaussElimination.h

Description: Gauss elimation to solve system of linear equations
Usage: variables are numbered 0 -> m-1, the m-th column is the right hand side
Time: $\mathcal{O}(min(N, M) \cdot N \cdot M)$ where N = equations, M = variables
Status: Tested on VNOI

9990b6, 38 lines

```
int gauss(vector<vector<double>> &a, vector<double> &ans
    ) {
  const double EPS = 1e-9;
  int n = a.size(), m = a[0].size() - 1;

  vector<int> where(m, -1);
  for (int col = 0, row = 0; col < m && row < n; ++col)
      {
    int p = row;
    for (int i = row; i < n; ++i)
      if (abs(a[i][col]) > abs(a[p][col])) p = i;

    if (abs(a[p][col]) < EPS) continue;

    swap(a[row], a[p]);
    where[col] = row;

    for (int i = 0; i < n; ++i) if (i != row) {
      double c = a[i][col] / a[row][col];
      for (int j = col; j <= m; ++j)
        a[i][j] -= a[row][j] * c;
    }
    ++row;
  }

  ans.assign(m, 0);
  for (int i = 0; i < m; ++i) if (where[i] != -1) {
    ans[i] = a[where[i]][m] / a[where[i]][i];
  }

  for (int i = 0; i < n; ++i) {
    double sum = 0;
    for (int j = 0; j < m; ++j) sum += ans[j] * a[i][j];
    if (abs(sum - a[i][m]) > EPS) return 0;
  }

  for (int i = 0; i < m; ++i)
    if (where[i] == -1) return 2;
  return 1;
}
```

GaussElimination2.h

Description: Gauss elimation to solve system of linear equations (mod 2)
Usage: all roots of the system have the form: ans + sum(c(i) * basis(i))
Time: $\mathcal{O}\left(\frac{min(N,M)\cdot N\cdot M}{64}\right)$ where N = equations, M = variables
Status: Tested on judge.yosupo.jp

1a50a3, 55 lines

```
namespace Gauss {
  // n: num equations, m: num variables
  // left hand side: [0, m-1], right hand side: column m
  const int M = 4096 + 5;
  using T = bitset<M>;

  int n, m;
  int where[M];
  vector<T> a;
  T ans;

  void init(int _n, int _m, const vector<T> &_a) {
    n = _n, m = _m, a = _a;
  }

  int gauss() {
    ans.reset();
    for (int i = 0; i < m; ++i) where[i] = -1;
    int row = 0;
    for (int col = 0; col < m && row < n; ++col) {
      for (int i = row; i < n; ++i) if (a[i][col]) {
        swap(a[i], a[row]);
        break;
      }
      if (!a[row][col]) continue;
      where[col] = row;
      for (int i = 0; i < n; ++i)
        if (i != row && a[i][col]) a[i] ^= a[row];
      ++row;
    }

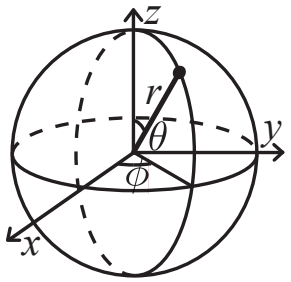
    for (int i = 0; i < m; ++i) if (where[i] != -1) {
      ans[i] = a[where[i]][m];
    }

    for (int i = 0; i < n; ++i) {
      bool sum = 0;
      for (int j = 0; j < m; ++j) sum ^= ans[j] & a[i][j
          ];
      if (sum != a[i][m]) return -1;
    }
    return row;
  }

  vector<T> get_basis() { // assume gauss() has been
      called
    vector<T> basis;
    for (int i = 0; i < m; ++i) if (where[i] == -1) {
```

```
      T x; x[i] = 1;
      // x(j) + a[where[j]][i] * x[i] = 0
      for (int j = 0; j < m; ++j)
        if (where[j] != -1) x[j] = a[where[j]][i];
      basis.emplace_back(x);
    }
    return basis;
  }
}
```

2.2.1 sphericalCoordinates



Combinatorial (3)

3.1 Permutations

3.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13	14	15	16	17			
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

3.1.2 Cycles

Let $g_S(n)$ be the number of n-permutations whose cycle lengths all belong to the set S. Then

$$\sum_{n=0}^{\infty} g_S(n)\frac{x^n}{n!} = \exp\left(\sum_{n\in S}\frac{x^n}{n}\right)$$

3.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1)+D(n-2)) = nD(n-1)+(-1)^n = \left\lfloor \frac{n!}{e} \right\rceil$$

3.1.4 Burnside's lemma

Given a group G of symmetries and a set X, the number of elements of X *up to symmetry* equals

$$\frac{1}{|G|}\sum_{g\in G}|X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts "configurations" (of some sort) of length n, we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n}\sum_{k=0}^{n-1} f(\gcd(n,k)) = \frac{1}{n}\sum_{k|n} f(k)\phi(n/k).$$

3.2 Partitions and subsets

3.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1,\ p(n) = \sum_{k\in\mathbb{Z}\setminus\{0\}} (-1)^{k+1} p(n-k(3k-1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	∼2e5	∼2e8

3.2.2 Lucas' Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + ... + n_1 p + n_0$ and $m = m_k p^k + ... + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^{k}\binom{n_i}{m_i} \pmod p$.

3.2.3 Binomials

3.3 General purpose numbers

3.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t-1}$ (FFT-able).
$B[0,\ldots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \ldots]$

Sums of powers:

$$\sum_{i=1}^{n} i^m = \frac{1}{m+1}\sum_{k=0}^{m}\binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^{\infty} f(i) = \int_m^{\infty} f(x)dx - \sum_{k=1}^{\infty}\frac{B_k}{k!}f^{(k-1)}(m)$$

$$\approx \int_m^{\infty} f(x)dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

3.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),\ c(0,0) = 1$$
$$\textstyle\sum_{k=0}^{n} c(n,k)x^k = x(x+1)\ldots(x+n-1)$$

$c(8,k) = 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$
$c(n,2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \ldots$

3.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j:s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j:s s.t. $\pi(j) \geq j$, k j:s s.t. $\pi(j) > j$.

$$E(n,k) = (n-k)E(n-1,k-1) + (k+1)E(n-1,k)$$

$$E(n,0) = E(n,n-1) = 1$$

$$E(n,k) = \sum_{j=0}^{k}(-1)^j\binom{n+1}{j}(k+1-j)^n$$

3.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$

$$S(n,1) = S(n,n) = 1$$

$$S(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}j^n$$

3.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \ldots.$ For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod p$$

3.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i: $n_1 n_2 \cdots n_k n^{k-2}$
with degrees d_i: $(n-2)!/((d_1-1)!\cdots(d_n-1)!)$

3.3.7 Catalan numbers

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1,\ C_{n+1} = \frac{2(2n+1)}{n+2}C_n,\ C_{n+1} = \sum C_i C_{n-i}$$

$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \ldots$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Number theory (4)

4.1 Modular arithmetic

ModularArithmetic.h

Description: Operators for modular arithmetic. You need to set mod to some number first and then you can use the structure.

"euclid.h" 35bfea, 18 lines

```
const ll mod = 17; // change to something else
struct Mod {
  ll x;
  Mod(ll xx) : x(xx) {}
  Mod operator+(Mod b) { return Mod((x + b.x) % mod); }
  Mod operator-(Mod b) { return Mod((x - b.x + mod) %
      mod); }
  Mod operator*(Mod b) { return Mod((x * b.x) % mod); }
  Mod operator/(Mod b) { return *this * invert(b); }
  Mod invert(Mod a) {
    ll x, y, g = euclid(a.x, mod, x, y);
    assert(g == 1); return Mod((x + mod) % mod);
  }
  Mod operator^(ll e) {
    if (!e) return Mod(1);
    Mod r = *this ^ (e / 2); r = r * r;
    return e&1 ? *this * r : r;
  }
};
```

ModInverse.h

Description: Pre-computation of modular inverses. Assumes LIM $\leq$ mod and that mod is a prime.

Status: Works

db454d, 3 lines

```
const ll mod = 1000000007, LIM = 200000;
vector<ll> inv(LIM); inv[1] = 1;
rep(i,2,LIM) inv[i] = mod - (mod / i) * inv[mod % i] %
    mod;
```

ModSqrt.h

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).

Time: $\mathcal{O}\left(\log^2 p\right)$ worst case, $\mathcal{O}\left(\log p\right)$ for most p

Status: Tested for all a,p <= 10000

19dd12, 34 lines

```
namespace mod_sqrt_detail {
ll power(ll a, ll e, ll p) {
  ll res = 1;
  for (; e; e >>= 1, a = a * a % p)
    if (e & 1) res = res * a % p;
  return res;
}
}

ll sqrt(ll a, ll p) {
  a %= p; if (a < 0) a += p;
  if (p == 2) return a;
  if (a == 0) return 0;
  assert(mod_sqrt_detail::power(a, (p-1)/2, p) == 1); //
      else no solution
  if (p % 4 == 3) return mod_sqrt_detail::power(a, (p+1)
      /4, p);
  // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8
      == 5
  ll s = p - 1, n = 2;
  int r = 0, m;
  while (s % 2 == 0)
    ++r, s /= 2;
  while (mod_sqrt_detail::power(n, (p - 1) / 2, p) != p
      - 1) ++n;
  ll x = mod_sqrt_detail::power(a, (s + 1) / 2, p);
  ll b = mod_sqrt_detail::power(a, s, p), g =
      mod_sqrt_detail::power(n, s, p);
  for (;; r = m) {
    ll t = b;
    for (m = 0; m < r && t != 1; ++m)
      t = t * t % p;
    if (m == 0) return x;
    ll gs = mod_sqrt_detail::power(g, 1LL << (r - m - 1)
        , p);
    g = gs * gs % p;
    x = x * gs % p;
    b = b * g % p;
  }
}
```

4.2 Primality

FastEratosthenes.h

Description: Prime sieve for generating all primes smaller than LIM.

Time: LIM=1e9 $\approx$ 1.5s

Status: Stress-tested

6b2912, 20 lines

```
const int LIM = 1e6;
bitset<LIM> isPrime;
vi eratosthenes() {
  const int S = (int)round(sqrt(LIM)), R = LIM / 2;
  vi pr = {2}, sieve(S+1); pr.reserve(int(LIM/log(LIM)
      *1.1));
  vector<pii> cp;
  for (int i = 3; i <= S; i += 2) if (!sieve[i]) {
    cp.push_back({i, i * i / 2});
    for (int j = i * i; j <= S; j += 2 * i) sieve[j] =
        1;
  }
  for (int L = 1; L <= R; L += S) {
    array<bool, S> block{};
    for (auto &[p, idx] : cp)
      for (int i=idx; i < S+L; idx = (i+=p)) block[i-L]
          = 1;
    rep(i,0,min(S, R - L))
      if (!block[i]) pr.push_back((L + i) * 2 + 1);
  }
  for (int i : pr) isPrime[i] = 1;
  return pr;
}
```

MillerRabin.h

Description: Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to $7 \cdot 10^{18}$; for larger numbers, use Python and extend A randomly.

Time: 7 times the complexity of $a^b \mod c$.

Status: Stress-tested

"ModMul.h" 60dcd1, 12 lines

```
bool isPrime(ull n) {
  if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
  ull A[] = {2, 325, 9375, 28178, 450775, 9780504,
      1795265022},
    s = __builtin_ctzll(n-1), d = n >> s;
  for (ull a : A) {   // ^ count trailing zeroes
    ull p = modpow(a%n, d, n), i = s;
    while (p != 1 && p != n - 1 && a % n && i--)
      p = modmul(p, p, n);
    if (p != n-1 && i != s) return 0;
  }
  return 1;
}
```

PollardRho.h

Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

Time: $\mathcal{O}\left(n^{1/4}\right)$, less for numbers with small factors.

Status: stress-tested

"ModMul.h", "MillerRabin.h" d8d98d, 18 lines

```
ull pollard(ull n) {
  ull x = 0, y = 0, t = 30, prd = 2, i = 1, q;
  auto f = [&](ull x) { return modmul(x, x, n) + i; };
  while (t++ % 40 || __gcd(prd, n) == 1) {
    if (x == y) x = ++i, y = f(x);
    if ((q = modmul(prd, max(x,y) - min(x,y), n))) prd =
        q;
    x = f(x), y = f(f(y));
  }
```

```
  return __gcd(prd, n);
}
vector<ull> factor(ull n) {
  if (n == 1) return {};
  if (isPrime(n)) return {n};
  ull x = pollard(n);
  auto l = factor(x), r = factor(n / x);
  l.insert(l.end(), all(r));
  return l;
}
```

4.3 Divisibility

euclid.h

Description: Finds two integers x and y, such that $ax + by = \gcd(a, b)$. If you just need gcd, use the built in __gcd instead. If a and b are coprime, then x is the inverse of $a \pmod b$.

33ba8f, 5 lines

```
ll euclid(ll a, ll b, ll &x, ll &y) {
  if (!b) return x = 1, y = 0, a;
  ll d = euclid(b, a % b, y, x);
  return y -= a/b * x, d;
}
```

Euclid.java

Description: Finds {x, y, d} s.t. ax + by = d = gcd(a, b).

Status: Tested

6aba01, 11 lines

```
static BigInteger[] euclid(BigInteger a, BigInteger b) {
  BigInteger x = BigInteger.ONE, yy = x;
  BigInteger y = BigInteger.ZERO, xx = y;
  while (b.signum() != 0) {
    BigInteger q = a.divide(b), t = b;
    b = a.mod(b); a = t;
    t = xx; xx = x.subtract(q.multiply(xx)); x = t;
    t = yy; yy = y.subtract(q.multiply(yy)); y = t;
  }
  return new BigInteger[]{x, y, a};
}
```

CRT.h

Description: Chinese Remainder Theorem.

crt(a, m, b, n) computes x such that $x \equiv a \pmod m$, $x \equiv b \pmod n$. If $|a| < m$ and $|b| < n$, x will obey $0 \le x < \text{lcm}(m, n)$. Assumes $mn < 2^{62}$.

Time: $\log(n)$

Status: Works

"euclid.h" 04d93a, 7 lines

```
ll crt(ll a, ll m, ll b, ll n) {
  if (n > m) swap(a, b), swap(m, n);
  ll x, y, g = euclid(m, n, x, y);
  assert((a - b) % g == 0); // else no solution
  x = (b - a) % n * x % n / g * m + a;
  return x < 0 ? x + m*n/g : x;
}
```

4.3.1 Bézout's identity

For $a \neq 0$, $b \neq 0$, then $d = gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

phiFunction.h

Description: *Euler's* ϕ function is defined as $\phi(n) :=$ # of positive integers $\le n$ that are coprime with n. $\phi(1) = 1$, p prime $\Rightarrow \phi(p^k) = (p-1)p^{k-1}$, m, n coprime $\Rightarrow \phi(mn) = \phi(m)\phi(n)$. If $n = p_1^{k_1} p_2^{k_2} ... p_r^{k_r}$ then $\phi(n) = (p_1 - 1)p_1^{k_1-1} ... (p_r - 1)p_r^{k_r-1}$. $\phi(n) = n \cdot \prod_{p|n}(1 - 1/p)$. $\sum_{d|n} \phi(d) = n$, $\sum_{1 \le k \le n, \gcd(k,n)=1} k = n\phi(n)/2, n > 1$

Euler's thm: a, n coprime $\Rightarrow a^{\phi(n)} \equiv 1 \pmod n$.

Fermat's little thm: p prime $\Rightarrow a^{p-1} \equiv 1 \pmod p$ $\forall a$.

Status: Tested

cf7d6d, 8 lines

```
const int LIM = 5000000;
int phi[LIM];

void calculatePhi() {
  rep(i,0,LIM) phi[i] = i&1 ? i : i/2;
  for (int i = 3; i < LIM; i += 2) if(phi[i] == i)
    for (int j = i; j < LIM; j += i) phi[j] -= phi[j] /
        i;
}
```

4.4 Fractions

ContinuedFractions.h

Description: Given N and a real number $x \ge 0$, finds the closest rational approximation p/q with $p, q \le N$. It will obey $|p/q - x| \le 1/qN$. For consecutive convergents, $p_{k+1}q_k - q_{k+1}p_k = (-1)^k$. ($p_k/q_k$ alternates between $> x$ and $< x$.) If x is rational, y eventually becomes ∞; if x is the root of a degree 2 polynomial the a's eventually become cyclic.

Time: $\mathcal{O}(\log N)$

Status: stress-tested for n <= 300

dd6c5e, 21 lines

```
typedef double d; // for N ~ 1e7; long double for N ~ 1
    e9
pair<ll, ll> approximate(d x, ll N) {
  ll LP = 0, LQ = 1, P = 1, Q = 0, inf = LLONG_MAX; d y
      = x;
  for (;;) {
    ll lim = min(P ? (N-LP) / P : inf, Q ? (N-LQ) / Q :
        inf),
      a = (ll)floor(y), b = min(a, lim),
      NP = b*P + LP, NQ = b*Q + LQ;
    if (a > b) {
      // If b > a/2, we have a semi-convergent that
          gives us a
      // better approximation; if b = a/2, we *may* have
          one.
      // Return {P, Q} here for a more canonical
          approximation.
      return (abs(x - (d)NP / (d)NQ) < abs(x - (d)P / (d
          )Q)) ?
        make_pair(NP, NQ) : make_pair(P, Q);
    }
    if (abs(y = 1/(y - (d)a)) > 3*N) {
      return {NP, NQ};
    }
    LP = P; P = NP;
    LQ = Q; Q = NQ;
  }
}
```

FracBinarySearch.h

Description: Given f and N, finds the smallest fraction $p/q \in [0, 1]$ such that $f(p/q)$ is true, and $p, q \le N$. You may want to throw an exception from f if it finds an exact solution, in which case N can be removed.

Usage: fracBS([](Frac f) { return f.p>=3*f.q; }, 10); // {1,3}

Time: $\mathcal{O}(\log(N))$

Status: stress-tested for n <= 300

27ab3e, 25 lines

```
struct Frac { ll p, q; };

template<class F>
Frac fracBS(F f, ll N) {
  bool dir = 1, A = 1, B = 1;
  Frac lo{0, 1}, hi{1, 1}; // Set hi to 1/0 to search
      (0, N]
  if (f(lo)) return lo;
  assert(f(hi));
  while (A || B) {
    ll adv = 0, step = 1; // move hi if dir, else lo
    for (int si = 0; step; (step *= 2) >>= si) {
      adv += step;
      Frac mid{lo.p * adv + hi.p, lo.q * adv + hi.q};
      if (abs(mid.p) > N || mid.q > N || dir == !f(mid))
          {
        adv -= step; si = 2;
      }
    }
    hi.p += lo.p * adv;
    hi.q += lo.q * adv;
    dir = !dir;
    swap(lo, hi);
    A = B; B = !!adv;
  }
  return dir ? hi : lo;
}
```

4.5 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2),\ \ b = k \cdot (2mn),\ \ c = k \cdot (m^2 + n^2),$$

with $m > n > 0$, $k > 0$, $m \perp n$, and either m or n even.

4.6 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a, except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

4.7 Estimates

$\sum_{d|n} d = O(n \log \log n)$.

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

4.8 Mobius Function

$$\mu(n) = \begin{cases} 0 & n \text{ is not square free} \\ 1 & n \text{ has even number of prime factors} \\ -1 & n \text{ has odd number of prime factors} \end{cases}$$

Mobius Inversion:

$$g(n) = \sum_{d|n} f(d) \Leftrightarrow f(n) = \sum_{d|n} \mu(d) g(n/d)$$

Other useful formulas/forms:

$\sum_{d|n} \mu(d) = [n = 1]$ (very useful)

$g(n) = \sum_{n|d} f(d) \Leftrightarrow f(n) = \sum_{n|d} \mu(d/n) g(d)$

$g(n) = \sum_{1 \le m \le n} f(\lfloor \frac{n}{m} \rfloor) \Leftrightarrow f(n) = \sum_{1 \le m \le n} \mu(m) g(\lfloor \frac{n}{m} \rfloor)$

Eratosthenes.h

Description: Prime sieve for generating all primes up to a certain limit. isprime[i] is true iff i is a prime.

Time: lim=100'000'000 $\approx$ 0.8 s. Runs 30% faster if only odd indices are stored.

Status: Tested

7c144c, 11 lines

```
const int MAX_PR = 5'000'000;
bitset<MAX_PR> isprime;
vi eratosthenesSieve(int lim) {
  isprime.set(); isprime[0] = isprime[1] = 0;
  for (int i = 4; i < lim; i += 2) isprime[i] = 0;
  for (int i = 3; i*i < lim; i += 2) if (isprime[i])
    for (int j = i*i; j < lim; j += i*2) isprime[j] = 0;
  vi pr;
  rep(i,2,lim) if (isprime[i]) pr.push_back(i);
  return pr;
}
```

Numerical (5)

5.1 Polynomials and recurrences

Polynomial.h

c9b7b0, 17 lines

```
struct Poly {
  vector<double> a;
  double operator()(double x) const {
    double val = 0;
    for (int i = sz(a); i--;) (val *= x) += a[i];
    return val;
  }
  void diff() {
    rep(i,1,sz(a)) a[i-1] = i*a[i];
    a.pop_back();
  }
  void divroot(double x0) {
    double b = a.back(), c; a.back() = 0;
    for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+
        b, b=c;
    a.pop_back();
  }
};
```

PolyRoots.h

Description: Finds the real roots to a polynomial.

Usage: `polyRoots({{2,-3,1}},-1e9,1e9) // solve x^2-3x+2 = 0`

Time: $\mathcal{O}\left(n^2 \log(1/\epsilon)\right)$

"Polynomial.h" b00bfe, 23 lines

```
vector<double> polyRoots(Poly p, double xmin, double
    xmax) {
  if (sz(p.a) == 2) { return {-p.a[0]/p.a[1]}; }
  vector<double> ret;
  Poly der = p;
  der.diff();
  auto dr = polyRoots(der, xmin, xmax);
  dr.push_back(xmin-1);
  dr.push_back(xmax+1);
  sort(all(dr));
  rep(i,0,sz(dr)-1) {
    double l = dr[i], h = dr[i+1];
    bool sign = p(l) > 0;
    if (sign ^ (p(h) > 0)) {
      rep(it,0,60) { // while (h - l > 1e-8)
        double m = (l + h) / 2, f = p(m);
        if ((f <= 0) ^ sign) l = m;
        else h = m;
      }
      ret.push_back((l + h) / 2);
    }
  }
  return ret;
}
```

PolyInterpolate.h

Description: Given n points (x[i], y[i]), computes an n-1-degree polynomial p that passes through them: $p(x) = a[0] * x^0 + ... + a[n-1] * x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi), k = 0 \ldots n-1$.

Time: $\mathcal{O}\left(n^2\right)$

08bf48, 13 lines

```
typedef vector<double> vd;
vd interpolate(vd x, vd y, int n) {
  vd res(n), temp(n);
  rep(k,0,n-1) rep(i,k+1,n)
    y[i] = (y[i] - y[k]) / (x[i] - x[k]);
  double last = 0; temp[0] = 1;
  rep(k,0,n) rep(i,0,n) {
    res[i] += y[k] * temp[i];
    swap(last, temp[i]);
    temp[i] -= last * x[k];
  }
  return res;
}
```

BerlekampMassey.h

Description: Recovers any n-order linear recurrence relation from the first $2n$ terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size $\le n$.

Usage: `berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}`

Time: $\mathcal{O}\left(N^2\right)$

Status: bruteforce-tested mod 5 for n <= 5 and all s

"../number-theory/ModPow.h" 96548b, 20 lines

```
vector<ll> berlekampMassey(vector<ll> s) {
  int n = sz(s), L = 0, m = 0;
  vector<ll> C(n), B(n), T;
  C[0] = B[0] = 1;

  ll b = 1;
  rep(i,0,n) { ++m;
    ll d = s[i] % mod;
    rep(j,1,L+1) d = (d + C[j] * s[i - j]) % mod;
    if (!d) continue;
    T = C; ll coef = d * modpow(b, mod-2) % mod;
    rep(j,m,n) C[j] = (C[j] - coef * B[j - m]) % mod;
    if (2 * L > i) continue;
    L = i + 1 - L; B = T; b = d; m = 0;
  }

  C.resize(L + 1); C.erase(C.begin());
```

```
  for (ll& x : C) x = (mod - x) % mod;
  return C;
}
```

LinearRecurrence.h

Description: Generates the k'th term of an n-order linear recurrence $S[i] = \sum_j S[i-j-1]tr[j]$, given $S[0 \dots \geq n-1]$ and $tr[0 \dots n-1]$. Faster than matrix multiplication. Useful together with Berlekamp–Massey.
Usage: `linearRec({0, 1}, {1, 1}, k) // k'th Fibonacci number`
Time: $\mathcal{O}\left(n^2 \log k\right)$
Status: bruteforce-tested mod 5 for n <= 5

f4e444, 26 lines

```
typedef vector<ll> Poly;
ll linearRec(Poly S, Poly tr, ll k) {
  int n = sz(tr);

  auto combine = [&](Poly a, Poly b) {
    Poly res(n * 2 + 1);
    rep(i,0,n+1) rep(j,0,n+1)
      res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
    for (int i = 2 * n; i > n; --i) rep(j,0,n)
      res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j])
          % mod;
    res.resize(n + 1);
    return res;
  };

  Poly pol(n + 1), e(pol);
  pol[0] = e[1] = 1;

  for (++k; k; k /= 2) {
    if (k % 2) pol = combine(pol, e);
    e = combine(e, e);
  }

  ll res = 0;
  rep(i,0,n) res = (res + pol[i + 1] * S[i]) % mod;
  return res;
}
```

5.2 Optimization

GoldenSectionSearch.h

Description: Finds the argument minimizing the function f in the interval $[a,b]$ assuming f is unimodal on the interval, i.e. has only one local minimum and no local maximum. The maximum error in the result is *eps*. Works equally well for maximization with a small change in the code. See TernarySearch.h in the Various chapter for a discrete version.
Usage: `double func(double x) { return 4+x+.3*x*x; }`
`double xmin = gss(-1000,1000,func);`
Time: $\mathcal{O}\left(\log((b-a)/\epsilon)\right)$
Status: tested

31d45b, 14 lines

```
double gss(double a, double b, double (*f)(double)) {
  double r = (sqrt(5)-1)/2, eps = 1e-7;
  double x1 = b - r*(b-a), x2 = a + r*(b-a);
  double f1 = f(x1), f2 = f(x2);
  while (b-a > eps)
    if (f1 < f2) { //change to > to find maximum
      b = x2; x2 = x1; f2 = f1;
      x1 = b - r*(b-a); f1 = f(x1);
    } else {
      a = x1; x1 = x2; f1 = f2;
      x2 = a + r*(b-a); f2 = f(x2);
    }
  return a;
}
```

Simplex.h

Description: Solves a general linear maximization problem: maximize $c^T x$ subject to $Ax \leq b$, $x \geq 0$. Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of $c^T x$ otherwise. The input vector is set to an optimal x (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that $x = 0$ is viable.
Usage: `vvd A = {{1,-1}, {-1,1}, {-1,-2}};`
`vd b = {1,1,-4}, c = {-1,-1}, x;`
`T val = LPSolver(A, b, c).solve(x);`
Time: $\mathcal{O}\left(NM * \#pivots\right)$, where a pivot may be e.g. an edge relaxation. $\mathcal{O}\left(2^n\right)$ in the general case.
Status: seems to work?

aa8530, 68 lines

```
typedef double T; // long double, Rational, double + mod <P>...
typedef vector<T> vd;
typedef vector<vd> vvd;

const T eps = 1e-8, inf = 1/.0;
#define MP make_pair
#define ltj(X) if(s == -1 || MP(X[j],N[j]) < MP(X[s],N[s])) s=j

struct LPSolver {
  int m, n;
  vi N, B;
  vvd D;

  LPSolver(const vvd& A, const vd& b, const vd& c) :
    m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
      rep(i,0,m) rep(j,0,n) D[i][j] = A[i][j];
      rep(i,0,m) { B[i] = n+i; D[i][n] = -1; D[i][n+1] = b[i];}
      rep(j,0,n) { N[j] = j; D[m][j] = -c[j]; }
      N[n] = -1; D[m+1][n] = 1;
    }

  void pivot(int r, int s) {
    T *a = D[r].data(), inv = 1 / a[s];
    rep(i,0,m+2) if (i != r && abs(D[i][s]) > eps) {
      T *b = D[i].data(), inv2 = b[s] * inv;
      rep(j,0,n+2) b[j] -= a[j] * inv2;
      b[s] = a[s] * inv2;
    }
    rep(j,0,n+2) if (j != s) D[r][j] *= inv;
    rep(i,0,m+2) if (i != r) D[i][s] *= -inv;
    D[r][s] = inv;
    swap(B[r], N[s]);
  }

  bool simplex(int phase) {
    int x = m + phase - 1;
    for (;;) {
      int s = -1;
      rep(j,0,n+1) if (N[j] != -phase) ltj(D[x]);
      if (D[x][s] >= -eps) return true;
      int r = -1;
      rep(i,0,m) {
        if (D[i][s] <= eps) continue;
        if (r == -1 || MP(D[i][n+1] / D[i][s], B[i])
              < MP(D[r][n+1] / D[r][s], B[r])) r = i;
      }
      if (r == -1) return false;
      pivot(r, s);
    }
  }

  T solve(vd &x) {
    int r = 0;
    rep(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
    if (D[r][n+1] < -eps) {
      pivot(r, n);
      if (!simplex(2) || D[m+1][n+1] < -eps) return -inf;
      rep(i,0,m) if (B[i] == -1) {
        int s = 0;
        rep(j,1,n+1) ltj(D[i]);
        pivot(i, s);
      }
    }
    bool ok = simplex(1); x = vd(n);
    rep(i,0,m) if (B[i] < n) x[B[i]] = D[i][n+1];
    return ok ? D[m][n+1] : inf;
  }
};
```

5.3 Matrices

Determinant.h

Description: Calculates determinant of a matrix. Destroys the matrix.
Time: $\mathcal{O}\left(N^3\right)$
Status: somewhat tested

bd5cec, 15 lines

```
double det(vector<vector<double>>& a) {
  int n = sz(a); double res = 1;
  rep(i,0,n) {
    int b = i;
    rep(j,i+1,n) if (fabs(a[j][i]) > fabs(a[b][i])) b = j;
    if (i != b) swap(a[i], a[b]), res *= -1;
    res *= a[i][i];
```

```
    if (res == 0) return 0;
    rep(j,i+1,n) {
      double v = a[j][i] / a[i][i];
      if (v != 0) rep(k,i+1,n) a[j][k] -= v * a[i][k];
    }
  }
  return res;
}
```

IntDeterminant.h

Description: Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.
Time: $\mathcal{O}\left(N^3\right)$
Status: bruteforce-tested for N <= 3, mod <= 7

3313dc, 18 lines

```
const ll mod = 12345;
ll det(vector<vector<ll>>& a) {
  int n = sz(a); ll ans = 1;
  rep(i,0,n) {
    rep(j,i+1,n) {
      while (a[j][i] != 0) { // gcd step
        ll t = a[i][i] / a[j][i];
        if (t) rep(k,i,n)
          a[i][k] = (a[i][k] - a[j][k] * t) % mod;
        swap(a[i], a[j]);
        ans *= -1;
      }
    }
    ans = ans * a[i][i] % mod;
    if (!ans) return 0;
  }
  return (ans + mod) % mod;
}
```

SolveLinear.h

Description: Solves $A * x = b$. If there are multiple solutions, an arbitrary one is returned. Returns rank, or -1 if no solutions. Data in A and b is lost.
Time: $\mathcal{O}\left(n^2m\right)$
Status: tested on kattis:equationsolver, and bruteforce-tested mod 3 and 5 for n,m <= 3

44c9ab, 38 lines

```
typedef vector<double> vd;
const double eps = 1e-12;

int solveLinear(vector<vd>& A, vd& b, vd& x) {
  int n = sz(A), m = sz(x), rank = 0, br, bc;
  if (n) assert(sz(A[0]) == m);
  vi col(m); iota(all(col), 0);

  rep(i,0,n) {
    double v, bv = 0;
    rep(r,i,n) rep(c,i,m)
      if ((v = fabs(A[r][c])) > bv)
        br = r, bc = c, bv = v;
    if (bv <= eps) {
      rep(j,i,n) if (fabs(b[j]) > eps) return -1;
      break;
    }
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    rep(j,0,n) swap(A[j][i], A[j][bc]);
    bv = 1/A[i][i];
    rep(j,i+1,n) {
      double fac = A[j][i] * bv;
      b[j] -= fac * b[i];
      rep(k,i+1,m) A[j][k] -= fac*A[i][k];
    }
    rank++;
  }

  x.assign(m, 0);
  for (int i = rank; i--;) {
    b[i] /= A[i][i];
    x[col[i]] = b[i];
    rep(j,0,i) b[j] -= A[j][i] * b[i];
  }
  return rank; // (multiple solutions if rank < m)
}
```

SolveLinear2.h

Description: To get all uniquely determined values of x back from SolveLinear, make the following changes:
Status: tested on kattis:equationsolverplus, stress-tested

"SolveLinear.h" 08e495, 7 lines

```
rep(j,0,n) if (j != i) // instead of rep(j,i+1,n)
// ... then at the end:
x.assign(m, undefined);
rep(i,0,rank) {
  rep(j,rank,m) if (fabs(A[i][j]) > eps) goto fail;
  x[col[i]] = b[i] / A[i][i];
fail:; }
```

SolveLinearBinary.h

Description: Solves $Ax = b$ over $\mathbb{F}_2$. If there are multiple solutions, one is returned arbitrarily. Returns rank, or -1 if no solutions. Destroys A and b.
Time: $\mathcal{O}\left(n^2m\right)$
Status: bruteforce-tested for n, m <= 4

fa2d7a, 34 lines

```
typedef bitset<1000> bs;

int solveLinear(vector<bs>& A, vi& b, bs& x, int m) {
  int n = sz(A), rank = 0, br;
  assert(m <= sz(x));
  vi col(m); iota(all(col), 0);
  rep(i,0,n) {
    for (br=i; br<n; ++br) if (A[br].any()) break;
    if (br == n) {
      rep(j,i,n) if(b[j]) return -1;
      break;
    }
    int bc = (int)A[br]._Find_next(i-1);
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    rep(j,0,n) if (A[j][i] != A[j][bc]) {
      A[j].flip(i); A[j].flip(bc);
    }
    rep(j,i+1,n) if (A[j][i]) {
      b[j] ^= b[i];
      A[j] ^= A[i];
    }
    rank++;
  }

  x = bs();
  for (int i = rank; i--;) {
    if (!b[i]) continue;
    x[col[i]] = 1;
    rep(j,0,i) b[j] ^= A[j][i];
  }
  return rank; // (multiple solutions if rank < m)
}
```

MatrixInverse.h

Description: Invert matrix A. Returns rank; result is stored in A unless singular (rank < n). Can easily be extended to prime moduli; for prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$ where A^{-1} starts as the inverse of A mod p, and k is doubled in each step.
Time: $\mathcal{O}\left(n^3\right)$
Status: Slightly tested

ebfff6, 35 lines

```
int matInv(vector<vector<double>>& A) {
  int n = sz(A); vi col(n);
  vector<vector<double>> tmp(n, vector<double>(n));
  rep(i,0,n) tmp[i][i] = 1, col[i] = i;

  rep(i,0,n) {
    int r = i, c = i;
    rep(j,i,n) rep(k,i,n)
      if (fabs(A[j][k]) > fabs(A[r][c]))
        r = j, c = k;
    if (fabs(A[r][c]) < 1e-12) return i;
    A[i].swap(A[r]); tmp[i].swap(tmp[r]);
    rep(j,0,n)
      swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c])
          ;
    swap(col[i], col[c]);
    double v = A[i][i];
    rep(j,i+1,n) {
      double f = A[j][i] / v;
      A[j][i] = 0;
      rep(k,i+1,n) A[j][k] -= f*A[i][k];
      rep(k,0,n) tmp[j][k] -= f*tmp[i][k];
    }
    rep(j,i+1,n) A[i][j] /= v;
    rep(j,0,n) tmp[i][j] /= v;
    A[i][i] = 1;
```

```
  }

  for (int i = n-1; i > 0; --i) rep(j,0,i) {
    double v = A[j][i];
    rep(k,0,n) tmp[j][k] -= v*tmp[i][k];
  }

  rep(i,0,n) rep(j,0,n) A[col[i]][col[j]] = tmp[i][j];
  return n;
}
```

MatrixInverse-mod.h

Description: Invert matrix A modulo a prime. Returns rank; result is stored in A unless singular (rank < n). For prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1})$ (mod p^k) where A^{-1} starts as the inverse of A mod p, and k is doubled in each step.
Time: $\mathcal{O}(n^3)$
Status: Slightly tested

"../number-theory/ModPow.h" 0b7b13, 37 lines

```
int matInv(vector<vector<ll>>& A) {
  int n = sz(A); vi col(n);
  vector<vector<ll>> tmp(n, vector<ll>(n));
  rep(i,0,n) tmp[i][i] = 1, col[i] = i;

  rep(i,0,n) {
    int r = i, c = i;
    rep(j,i,n) rep(k,i,n) if (A[j][k]) {
      r = j; c = k; goto found;
    }
    return i;
found:
    A[i].swap(A[r]); tmp[i].swap(tmp[r]);
    rep(j,0,n)
      swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c])
          ;
    swap(col[i], col[c]);
    ll v = modpow(A[i][i], mod - 2);
    rep(j,i+1,n) {
      ll f = A[j][i] * v % mod;
      A[j][i] = 0;
      rep(k,i+1,n) A[j][k] = (A[j][k] - f*A[i][k]) % mod
          ;
      rep(k,0,n) tmp[j][k] = (tmp[j][k] - f*tmp[i][k]) %
           mod;
    }
    rep(j,i+1,n) A[i][j] = A[i][j] * v % mod;
    rep(j,0,n) tmp[i][j] = tmp[i][j] * v % mod;
    A[i][i] = 1;
  }

  for (int i = n-1; i > 0; --i) rep(j,0,i) {
    ll v = A[j][i];
    rep(k,0,n) tmp[j][k] = (tmp[j][k] - v*tmp[i][k]) %
        mod;
  }
  rep(i,0,n) rep(j,0,n)
    A[col[i]][col[j]] = tmp[i][j] % mod + (tmp[i][j] <
        0)*mod;
  return n;
}
```

Tridiagonal.h

Description: $x = \text{tridiagonal}(d, p, q, b)$ solves the equation system

$$\begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_{n-1} \end{pmatrix} = \begin{pmatrix} d_0 & p_0 & 0 & 0 & \cdots & 0 \\ q_0 & d_1 & p_1 & 0 & \cdots & 0 \\ 0 & q_1 & d_2 & p_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & q_{n-3} & d_{n-2} & p_{n-2} \\ 0 & 0 & \cdots & 0 & q_{n-2} & d_{n-1} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \end{pmatrix}.$$

This is useful for solving problems on the type

$$a_i = b_i a_{i-1} + c_i a_{i+1} + d_i,\ 1 \le i \le n,$$

where a_0, a_{n+1}, b_i, c_i and d_i are known. a can then be obtained from

$$\{a_i\} = \text{tridiagonal}(\{1, -1, -1, ..., -1, 1\}, \{0, c_1, c_2, \ldots, c_n\}, \{b_1, b_2, \ldots, b_n, 0\}, \{a_0, d_1, d_2, \ldots, d_n, a_{n+1}\}).$$

Fails if the solution is not unique.
If $|d_i| > |p_i| + |q_{i-1}|$ for all i, or $|d_i| > |p_{i-1}| + |q_i|$, or the matrix is positive definite, the algorithm is numerically stable and neither `tr` nor the check for `diag[i] == 0` is needed.
Time: $\mathcal{O}(N)$
Status: Brute-force tested mod 5 and 7 and stress-tested for real matrices obeying the criteria above.

8f9fa8, 26 lines

```
typedef double T;
vector<T> tridiagonal(vector<T> diag, const vector<T>&
    super,
    const vector<T>& sub, vector<T> b) {
  int n = sz(b); vi tr(n);
  rep(i,0,n-1) {
    if (abs(diag[i]) < 1e-9 * abs(super[i])) { // diag[i
        ] == 0
      b[i+1] -= b[i] * diag[i+1] / super[i];
      if (i+2 < n) b[i+2] -= b[i] * sub[i+1] / super[i];
      diag[i+1] = sub[i]; tr[++i] = 1;
    } else {
      diag[i+1] -= super[i]*sub[i]/diag[i];
      b[i+1] -= b[i]*sub[i]/diag[i];
    }
  }
  for (int i = n; i--;) {
    if (tr[i]) {
      swap(b[i], b[i-1]);
      diag[i-1] = diag[i];
      b[i] /= super[i-1];
    } else {
      b[i] /= diag[i];
      if (i) b[i-1] -= b[i]*super[i-1];
    }
  }
  return b;
}
```

5.4 Fourier transforms

FastFourierTransform.h

Description: fft(a) computes $\hat{f}(k) = \sum_x a[x] \exp(2\pi i \cdot kx/N)$ for all k. N must be a power of 2. Useful for convolution: `conv(a, b) = c`, where $c[x] = \sum a[i]b[x-i]$. For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by n, reverse(start+1, end), FFT back. Rounding is safe if $(\sum a_i^2 + \sum b_i^2)\log_2 N < 9 \cdot 10^{14}$ (in practice 10^{16}; higher for random inputs). Otherwise, use NTT/FFTMod.
Time: $\mathcal{O}(N \log N)$ with $N = |A| + |B|$ ($\sim 1s$ for $N = 2^{22}$)
Status: somewhat tested

00ced6, 35 lines

```
typedef complex<double> C;
typedef vector<double> vd;
void fft(vector<C>& a) {
  int n = sz(a), L = 31 - __builtin_clz(n);
  static vector<complex<long double>> R(2, 1);
  static vector<C> rt(2, 1);  // (^ 10% faster if double
      )
  for (static int k = 2; k < n; k *= 2) {
    R.resize(n); rt.resize(n);
    auto x = polar(1.0L, acos(-1.0L) / k);
    rep(i,k,2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i
        /2];
  }
  vi rev(n);
  rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
  rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
  for (int k = 1; k < n; k *= 2)
    for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
      C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-
          rolled)
      a[i + j + k] = a[i + j] - z;
      a[i + j] += z;
    }
}
vd conv(const vd& a, const vd& b) {
  if (a.empty() || b.empty()) return {};
  vd res(sz(a) + sz(b) - 1);
  int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
  vector<C> in(n), out(n);
  copy(all(a), begin(in));
  rep(i,0,sz(b)) in[i].imag(b[i]);
  fft(in);
  for (C& x : in) x *= x;
  rep(i,0,n) out[i] = in[-i & (n - 1)] - conj(in[i]);
  fft(out);
  rep(i,0,sz(res)) res[i] = imag(out[i]) / (4 * n);
  return res;
}
```

FastFourierTransformMod.h

Description: Higher precision FFT, can be used for convolutions modulo arbitrary integers as long as $N \log_2 N \cdot \text{mod} < 8.6 \cdot 10^{14}$ (in practice 10^{16} or higher). Inputs must be in $[0, \text{mod})$.
Time: $\mathcal{O}(N \log N)$, where $N = |A| + |B|$ (twice as slow as NTT or FFT)
Status: stress-tested

"FastFourierTransform.h" b82773, 22 lines

```
typedef vector<ll> vl;
template<int M> vl convMod(const vl &a, const vl &b) {
  if (a.empty() || b.empty()) return {};
  vl res(sz(a) + sz(b) - 1);
  int B=32-__builtin_clz(sz(res)), n=1<<B, cut=int(sqrt(
      M));
  vector<C> L(n), R(n), outs(n), outl(n);
  rep(i,0,sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] %
      cut);
  rep(i,0,sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] %
      cut);
  fft(L), fft(R);
  rep(i,0,n) {
    int j = -i & (n - 1);
    outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
    outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1
        i;
  }
  fft(outl), fft(outs);
  rep(i,0,sz(res)) {
    ll av = ll(real(outl[i])+.5), cv = ll(imag(outs[i])
        +.5);
    ll bv = ll(imag(outl[i])+.5) + ll(real(outs[i])+.5);
    res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
  }
  return res;
}
```

NumberTheoreticTransform.h

Description: ntt(a) computes $\hat{f}(k) = \sum_x a[x] g^{xk}$ for all k, where $g = \text{root}^{(mod-1)/N}$. N must be a power of 2. Useful for convolution modulo specific nice primes of the form $2^a b + 1$, where the convolution result has size at most 2^a. For arbitrary modulo, see FFTMod. `conv(a, b) = c`, where $c[x] = \sum a[i] b[x-i]$. For manual convolution: NTT the inputs, multiply pointwise, divide by n, reverse(start+1, end), NTT back. Inputs must be in [0, mod).
Time: $\mathcal{O}(N \log N)$
Status: stress-tested

ccea62, 41 lines

```
const ll mod = (119 << 23) + 1, root = 62; // =
    998244353
ll nttPow(ll a, ll e) {
  ll res = 1;
  for (; e; e >>= 1, a = a * a % mod)
    if (e & 1) res = res * a % mod;
  return res;
}
// For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479
    << 21
// and 483 << 21 (same root). The last two are > 10^9.
typedef vector<ll> vl;
void ntt(vl &a) {
  int n = sz(a), L = 31 - __builtin_clz(n);
  static vl rt(2, 1);
  for (static int k = 2, s = 2; k < n; k *= 2, s++) {
    rt.resize(n);
    ll z[] = {1, nttPow(root, mod >> s)};
    rep(i,k,2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
  }
  vi rev(n);
  rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
  rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
  for (int k = 1; k < n; k *= 2)
    for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
      ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i +
          j];
      a[i + j + k] = ai - z + (z > ai ? mod : 0);
      ai += (ai + z >= mod ? z - mod : z);
    }
}
vl conv(const vl &a, const vl &b) {
  if (a.empty() || b.empty()) return {};
  int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s),
    n = 1 << B;
  int inv = nttPow(n, mod - 2);
  vl L(a), R(b), out(n);
  L.resize(n), R.resize(n);
  ntt(L), ntt(R);
  rep(i,0,n)
    out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv %
        mod;
  ntt(out);
  return {out.begin(), out.begin() + s};
}
```

FastSubsetTransform.h

Description: Transform to a basis with fast convolutions of the form $c[z] = \sum_{z=x \oplus y} a[x] \cdot b[y]$, where $\oplus$ is one of AND, OR, XOR. The size of a must be a power of two.
Time: $\mathcal{O}(N \log N)$
Status: stress-tested

464cf3, 16 lines

```
void FST(vi& a, bool inv) {
  for (int n = sz(a), step = 1; step < n; step *= 2) {
    for (int i = 0; i < n; i += 2 * step) rep(j,i,i+step
        ) {
      int &u = a[j], &v = a[j + step]; tie(u, v) =
        inv ? pii(v - u, u) : pii(v, u + v); // AND
        inv ? pii(v, u - v) : pii(u + v, u); // OR
        pii(u + v, u - v);                   // XOR
    }
  }
  if (inv) for (int& x : a) x /= sz(a); // XOR only
}
vi conv(vi a, vi b) {
  FST(a, 0); FST(b, 0);
  rep(i,0,sz(a)) a[i] *= b[i];
  FST(a, 1); return a;
}
```

Data structures (6)

OrderSet.h

Description: Ordered set.
Usage: `0-indexed.`
`find_by_order(x) returns x-th element (ascending order)`
`order_of_key(x) returns no. of elements <x.`
Time: $\mathcal{O}(\log N)$ for each operation
Status: Tested on one problem.

<ext/pb_ds/assoc_container.hpp> df383f, 3 lines

```
using namespace __gnu_pbds;
template <class T>
using orderset = tree<T, null_type, less<T>, rb_tree_tag
    , tree_order_statistics_node_update>;
```

FastSet.h

Description: Fast set for sets that only have integers in [1, N]
Usage: `call init() first`
Memory: $\mathcal{O}(N)$
Time: $\mathcal{O}(log64(n))$ for update and lowerbound, upperbound
Status: Well-tested

e52539, 92 lines

```
struct FastSet {
  static const int N = 250000 + 5;
  static const int BLOCK = 64;
  static const int LAYER = 3;
  uint64_t a[N + 5]{};
  int layer_start[LAYER];
  FastSet(){
    int start = 0, pos = 1;
    for(int i = 0; i < LAYER; ++i) {
      layer_start[i] = start;
      start += pos;
      pos = pos * BLOCK;
    }
  }
  void init() { fill(a, a + N + 5, 0); }

  void update(int x) {
    assert(1 <= x && x <= N);
    //add if x not in set and vice versa
    int id = x >> 6, prev = layer_start[LAYER - 1] + id;
    a[prev] ^= (1ULL << (x & (BLOCK - 1)));
    for (int i = LAYER - 2; i >= 0; --i) {
      x >>= 6, id >>= 6;
      int digit = x & (BLOCK - 1), cur = id +
          layer_start[i];
      if ((a[prev] > 0) != ((a[cur] >> digit) & 1ULL)) a
          [cur] ^= (1ULL << digit);
      prev = cur;
    }
```

```
}

int get_next(uint64_t mask, int l) {
  if (l >= BLOCK) return -1;
  mask &= ~uint64_t(0) << l;
  return mask ? __builtin_ctzll(mask) : -1;
}

int find_next(int x) { // return first ele >= x or -1
    if no ele
  int id = x >> 6;
  int foo = get_next(a[id + layer_start[LAYER - 1]], x
       & (BLOCK - 1));
  if (foo != -1) return x - (x & (BLOCK - 1)) + foo;
  x >>= 6, id >>= 6;
  for (int i = LAYER - 2; i>=0; --i) {
    int digit = (x & (BLOCK - 1)) + 1;
    int cur = layer_start[i] + id;
    int foo = get_next(a[cur], digit);
    if (foo != -1) {
      id = (id << 6) + foo;
      for (int j = i + 1; j < LAYER; ++j) {
        int digit = __builtin_ctzll(a[id + layer_start
            [j]]);
        id = (id << 6) + digit;
      }
      return id;
    }
    x >>= 6, id >>= 6;
  }
  return -1;
}

int get_prev(uint64_t mask, int l) {
  if (l < BLOCK) mask &= (1ULL << l) - 1;
  return mask ? 63 - __builtin_clzll(mask) : -1;
}

int find_prev(int x) { // return last ele < x or -1 if
     no ele
  int id = x >> 6;
  int foo = get_prev(a[id + layer_start[LAYER - 1]], x
       & (BLOCK - 1));
  if (foo != -1) return x - (x & (BLOCK - 1)) + foo;
  x >>= 6, id >>= 6;
  for(int i = LAYER - 2; i>=0; --i) {
    int digit = (x & (BLOCK - 1)) + 1;
    int cur = layer_start[i] + id;
    int foo = get_prev(a[cur], digit - 1);
    if (foo != -1) {
      id = (id << 6) + foo;
      for(int j = i + 1; j < LAYER; ++j) {
        int digit = (63 - __builtin_clzll(a[id +
            layer_start[j]]));
        id = (id << 6) + digit;
      }
      return id;
    }
    x >>= 6, id >>= 6;
  }
  return -1;
}

int get_min() {
  return find_next(1);
}

int get_max() {
  return find_prev(N + 1);
}
};
```

LichaoTree.h

Description: Li Chao Tree, can query min and add line.
Time: Both operations are $\mathcal{O}(\log N)$
Status: Tested by KickingKun

764eb9, 49 lines

```
struct LiChaoTree {
  //range for x is in [-lim, lim].
  static const ll lim = 1e9, INF = 1e18;
  struct Line {
    ll a, b;
    ll operator() (ll x) {
      return a * x + b;
    }
  };

  struct node {
    Line line;
    node *l, *r;
    node() {
      line = {0, INF};
      l = r = NULL;
    }
  };

  node *root = new node();

  ll query(node *i, ll l, ll r, ll x) {
    if (i == NULL || x < l || x > r) return INF;
    ll m = (l + r) >> 1;
    ll ans = i->line(x);
    ans = min(ans, query(i->l, l, m, x));
    ans = min(ans, query(i->r, m + 1, r, x));
    return ans;
  }

  ll query(ll x) {
    return query(root, -lim, lim, x);
  }

  void add(Line li, node *&cur, ll l, ll r) {
    if (cur == nullptr){
      cur = new node(); cur->line = li;
      return;
    }
    ll mid = (l + r) >> 1;
    if (li(mid) < cur->line(mid)) swap(li, cur->line);
    if (li(l) < cur->line(l)) add(li, cur->l, l, mid);
    if (li(r) < cur->line(r)) add(li, cur->r, mid + 1, r
        );
  }

  void add(ll m, ll b) {
    add({m, b}, root, -lim, lim);
  }
} cht;
```

PersistentIT.h

Description: Persistent segment tree with point add updates and range sum queries.
Usage: `Remember to call mytree.resize(n) before any operations.`
`Make sure to check for integer overflows in update values.`
Memory: $\mathcal{O}(\log N \cdot numUpdates)$
Time: $\mathcal{O}(\log N)$ for each operation
Status: Tested on CSES

c22699, 53 lines

```
//change the parameters
const int MAX_UPDATES = 1e7+10;
struct Node {
  int le, ri;
  ll sum;
  Node(int _le = 0, int _ri = 0, ll _sum = 0) {
    le = _le, ri = _ri, sum = _sum;
  }
} nodes[MAX_UPDATES];
struct PersistentIT { // 1-indexed array
  int n;
  PersistentIT(int _n = 0) {
    resize(_n);
  }

  void resize(int _n) {
    n = _n;
  }
  int numNode = 0;
  int update(int oldId, int l, int r, int p, int val) {
    if (l > p || r < p) return oldId;

    assert(numNode + 1 < MAX_UPDATES);
    numNode++; int id = numNode;
    if (l == r) {
      nodes[id] = nodes[oldId];
      nodes[id].sum += val;
      return id;
    }

    int mid = (l + r) >> 1;
```

```
    nodes[id].le = update(nodes[oldId].le, l, mid, p,
        val);
    nodes[id].ri = update(nodes[oldId].ri, mid + 1, r, p
        , val);

    nodes[id].sum = nodes[nodes[id].le].sum + nodes[
        nodes[id].ri].sum;
    return id;
  }

  int update(int root, int p, int val) {
    return update(root, 1, n, p, val);
  }

  ll get(int id, int l, int r, int u, int v) {
    if (l > v || r < u) return 0;
    if (u <= l && r <= v) return nodes[id].sum;
    int mid = (l + r) >> 1;
    return get(nodes[id].le, l, mid, u, v) + get(nodes[
        id].ri, mid + 1, r, u, v);
  }

  ll get(int root, int u, int v) {
    return get(root, 1, n, u, v);
  }
} mytree;
```

PersistentLazyIT.h

Description: Persistent segment tree with range updates and range sum queries.
Usage: `Remember to call mytree.resize(n) before any operations.`
Memory: $\mathcal{O}(\log N \cdot numUpdates)$
Time: $\mathcal{O}(\log N)$ amortized per update, $\mathcal{O}(\log N)$ per query
Status: Randomized version-history regression test

e5c3a6, 78 lines

```
struct Node {
  int le, ri;
  ll lazy, sum;

  Node(int _le = 0, int _ri = 0, ll _lazy = 0, ll _sum =
      0) {
    le = _le, ri = _ri, lazy = _lazy, sum = _sum;
  }
};
vector<Node> nodes(1); // Index 0 is the all-zero
    subtree.
int numNode = 0;
vector<int> version;

struct PersistentLazyIT {
  int n;

  PersistentLazyIT(int _n = 0) {
    resize(_n);
  }

  void resize(int _n) {
    n = _n;
  }

  int newlazykid(int oldId, int l, int r, ll val) {
    int id = ++numNode;
    nodes.push_back(nodes[oldId]);

    nodes[id].sum += 1ll * val * (r - l + 1);
    nodes[id].lazy += val;

    return id;
  }

  int newparent(int le, int ri) {
    int id = ++numNode;
    nodes.emplace_back();
    nodes[id].le = le, nodes[id].ri = ri;
    nodes[id].sum = nodes[le].sum + nodes[ri].sum;
    return id;
  }

  void push(int id, int l, int r) {
    if (nodes[id].lazy == 0) return;
    if (l != r) {
      int mid = (l + r) >> 1;
      nodes[id].le = newlazykid(nodes[id].le, l, mid,
          nodes[id].lazy);
      nodes[id].ri = newlazykid(nodes[id].ri, mid + 1, r
          , nodes[id].lazy);
    }
    nodes[id].lazy = 0;
  }

  int update(int oldId, int l, int r, int u, int v, ll
      val) {
    if (l > v || r < u) return oldId;
    if (u <= l && r <= v) return newlazykid(oldId, l, r,
        val);
    int copy = newlazykid(oldId, l, r, 0);
    push(copy, l, r); int mid = (l + r) >> 1;
    int le = update(nodes[copy].le, l, mid, u, v, val),
      ri = update(nodes[copy].ri, mid + 1, r, u, v, val)
          ;
    return newparent(le, ri);
  }

  int update(int root, int u, int v, ll val) {
    assert(n > 0);
    return update(root, 1, n, u, v, val);
  }

  ll get(int id, int l, int r, int u, int v, ll carry =
      0) {
    if (l > v || r < u) return 0;
    if (u <= l && r <= v) return nodes[id].sum + carry *
        (r - l + 1);
    carry += nodes[id].lazy; int mid = (l + r) >> 1;
    return get(nodes[id].le, l, mid, u, v, carry) + get(
        nodes[id].ri, mid + 1, r, u, v, carry);
  }

  ll get(int root, int u, int v) {
    assert(n > 0);
    return get(root, 1, n, u, v);
  }
} mytree;
```

ConvexHullTrickMonotonicSlopes.h

Description: Fast convex hull trick for monotonic slopes
Time: $\mathcal{O}(1)$ for addition and $\mathcal{O}(1)$ or $\mathcal{O}(\log N)$ for query
Status: Well-tested

6042b6, 63 lines

```
// Adding decreasing slopes, query min
// Use get() if query increasing, query() otherwise
struct ConvexHullTrick {
  const ll INF = 1e18;
  struct Line {
    ll m, b;

    Line(ll _m = 0, ll _b = 0) {
      m = _m, b = _b;
    }

    ll eval(ll x) {
      return m * x + b;
    }
  };

  bool isBad(Line l1, Line l2, Line l3) {
    // bad when x of (l1 x l3) < x of (l1 x l2)
    return ((__int128)l3.b - l1.b) * ((__int128)l1.m -
        l2.m)
      <= ((__int128)l2.b - l1.b) * ((__int128)l1.m - l3
          .m);
  }

  vector<Line> lines;
  int pointer = 0;

  ConvexHullTrick() {
    lines.clear(), pointer = 0;
  }

  void reset() {
    lines.clear(), pointer = 0;
  }

  void add(ll m, ll b) { // insert line decreasing in
      slope
    while (!lines.empty() && lines.back().m == m &&
        lines.back().b > b)
```

```
    lines.pop_back();
  if (!lines.empty() && lines.back().m == m) return;
  Line newLine(m, b);
  while (lines.size() > 1 && isBad(lines[lines.size()
      - 2], lines.back(), newLine))
    lines.pop_back();
  lines.emplace_back(newLine);
 }

 ll query(ll x) { // return min
  if (lines.empty()) return INF;
  if (lines.size() == 1) return lines[0].eval(x);
  int low = 0, high = int(lines.size()) - 2, best =
      -1;
  while (low <= high) {
    int mid = (low + high) >> 1;
    if (lines[mid + 1].eval(x) < lines[mid].eval(x))
      low = (best = mid) + 1;
    else high = mid - 1;
  }
  return lines[best + 1].eval(x);
 }

 ll get(ll x) { //answer query increasing in x
  if (lines.empty()) return INF;
  if (pointer >= lines.size()) pointer = lines.size()
      - 1;
  while (pointer + 1 < lines.size() && lines[pointer +
      1].eval(x) < lines[pointer].eval(x))
    ++pointer;
  return lines[pointer].eval(x);
 }
} cht;
```

FenwickTree2d.h

Description: Computes sums a[i,j] for all 0<=i<=I, 0<=j<=J, and increases single elements a[i,j] (0-index).
Usage: `Requires update positions to be known in advance in form vector<pair<int,int>> as constructor for FT2, or construct with limX only and call fakeUpdate().`
Time: $\mathcal{O}\left(\log^2 N\right)$.
Status: stress-tested

47b6a8, 55 lines

```
struct FT {
 vector<ll> s;
 FT(int n) : s(n) {}
 void update(int pos, ll dif) { // a[pos] += dif
  for (; pos < sz(s); pos |= pos + 1) s[pos] += dif;
 }
 ll query(int pos) { // sum of values in [0, pos)
  ll res = 0;
  for (; pos; pos -= (pos & (-pos))) res += s[pos -
      1];
  return res;
 }
 int lower_bound(ll sum) {// min pos st sum of [0, pos]
     >= sum
  // Returns n if no sum is >= sum, or -1 if empty sum
      is.
  if (sum <= 0) return -1;
  int pos = 0;
  for (int pw = 1 << 25; pw; pw >>= 1) {
   if (pos + pw <= sz(s) && s[pos + pw - 1] < sum)
    pos += pw, sum -= s[pos - 1];
  }
  return pos;
 }
};

struct FT2 {
 vector<vi> ypos; vector<FT> ybit;
 bool call_init = false;
 void fakeUpdate(int x, int y) {
  for (; x < sz(ypos); x |= x + 1) ypos[x].push_back(y
      );
 }
 FT2(int limx, vector<pair<int,int>> &updates){
  ypos.resize(limx + 5);
  for(auto [x, y]: updates) fakeUpdate(x, y);
  init();
 }
 FT2(int limx) : ypos(limx + 5) {}
 void init() {
  call_init = true;
  ybit.clear();
  for (vi& v : ypos) sort(all(v)), ybit.emplace_back(
      sz(v));
 }
 int get_id(int x, int y) {
  return (int)(lower_bound(all(ypos[x]), y) - ypos[x].
      begin());
 }
 void update(int x, int y, ll dif) {
  if(!call_init) init();
  for (; x < sz(ypos); x |= x + 1) ybit[x].update(
      get_id(x, y), dif);
 }
 ll query(int x, int y) { //calculates sum for <=x, <=y
     .
  if(!call_init) init();
  x++;
  ll sum = 0;
  for (; x; x -= (x & (-x))) sum += ybit[x - 1].query(
      upper_bound(all(ypos[x - 1]), y) - ypos[x - 1].
      begin());
  return sum;
 }
};
```

ImplicitTreap.h

Description: An implicit treap on an array that supports deleting an element, inserting an element in any position and splitting the array into two parts (prefix/suffix).
Usage: `Declare pnode t = nullptr for an empty array, then use the functions. If you want to handle more data in each node, then reimplement refresh(). Note that the array is 1-indexed.`
Time: $\mathcal{O}\left(\log N\right)$
Status: Tested on one problem

27f8be, 39 lines

```
mt19937_64 rng(6736);
struct node{
 int val, sz = 1;
 uint64_t y;
 node *le = nullptr, *ri = nullptr;
 node(int val): val(val), y(rng()) {}
};
namespace implicit_treap{ //1-index
 using pnode = node*;
 int get_sz(pnode &p){ return p != nullptr ? p->sz : 0;
     }
 void refresh(pnode &p){ p->sz = get_sz(p->le) + get_sz
     (p->ri) + 1; }
 //first is [1, i], second is [i + 1, n]
 pair<pnode, pnode> split(pnode t, int i){
  if(t == nullptr) return {nullptr, nullptr};
  pair<pnode, pnode> ans;
  if(get_sz(t->le) + 1 <= i){
   ans = split(t->ri, i - get_sz(t->le) - 1);
   t->ri = ans.first; ans.first = t;
   refresh(ans.first);
  }
  else{
   ans = split(t->le, i);
   t->le = ans.second; ans.second = t;
   refresh(ans.second);
  }
  return ans;
 }
 //t1 is treap of a1,...,an. t2 is treap of b1,...,bm.
 //Merge into a1,...,an,b1,...,bm.
 pnode merge(pnode t1, pnode t2){
  if(t1 == nullptr || t2 == nullptr) return t1 ==
      nullptr ? t2 : t1;
  if(t1->y >= t2->y){
   t1->ri = merge(t1->ri, t2);
   refresh(t1); return t1;
  }
  t2->le = merge(t1, t2->le);
  refresh(t2); return t2;
 }
}
```

Matrix.h

Description: Matrix multiplication modulo MOD. Stored entries must be in [0, MOD).
Usage: `Set MOD; 8 * MOD * MOD must fit in signed 64-bit.`
Status: Randomized multiplication and exponentiation regression test

14a4ba, 25 lines

```
struct Matrix {
  static const ll MOD = 1000000007; // Set to the
      required modulus.
  static const uint64_t modmod = 8ULL * MOD * MOD; //
      must be < 2^63
  vector<vector<uint64_t>> val;
  int n, m;

  void resize(int _n, int _m) {
    n = _n, m = _m;
    val.assign(n, vector<uint64_t>(m));
  }
  Matrix operator * (const Matrix &other) {
    Matrix ret(n, other.m);
    for (int i = 0; i < n; ++i)
      for (int k = 0; k < m; ++k) {
        for (int j = 0; j < other.m; ++j)
          ret[i][j] += val[i][k] * other[k][j];
        if (k % 8 == 0) {
          for (int j = 0; j < other.m; ++j)
            ret[i][j] = min(ret[i][j], ret[i][j] -
                modmod);
        }
      }
    for (auto &row : ret.val) for (auto &x : row) x %=
        MOD;
    return ret;
  }
};
```

MoQueries.h

Description: Using Mo's Algorithm to answer tree path (vertex) or interval queries, adding/removing one element each step.
Usage: change block size and Type of ans in t_ans and blsz.
Define add, remove and calc in namespace MoAlgo after including, then call mo_inter() or mo_tree().
Note that the the array/tree is 0-index.
Time: $\mathcal{O}\left(N\sqrt{Q}\right)$
Status: Tested on one tree problem and one interval problem.
4ea09c, 61 lines

```
namespace MoAlgo{
  using T_ans = ll;
  const int blsz = 350; // block size
  //(inter[0], inter[1]) is the current interval/path
      BEFORE OPERATION
  void add(int idx, array<int, 2> &inter); // add idx
  void remove(int idx, array<int, 2> &inter); // remove
      idx
  T_ans calc(array<int, 2> &inter);
  // compute current answer

  pii get_block(pii que){
    if((que.first / blsz) & 1) que.second = -que.second;
    return {que.first / blsz, que.second};
  }
  inline vector<T_ans> mo_tree(vector<pii> que, vector<
      vi>& graph, int root=0){
    int n = sz(graph);
    array<int, 2> path = {root, root};
    vi order(sz(que)), in(n), out(n), on_path(n), pa(n);
    vector<T_ans> ans(sz(que));
    int tim = 0;
    add(root, path), on_path[root] = 1;
    auto dfs = [&](int u, int p, auto& self) -> void {
      pa[u] = p;
      in[u] = tim; tim++;
      for (int v : graph[u]) if (v != p) self(v, u, self
          );
      out[u] = tim; tim++;
    };
    dfs(root, -1, dfs);
    iota(all(order), 0);
    auto cmp = [&](int s, int t){
      pii ps = {in[que[s].first], in[que[s].second]};
      pii pt = {in[que[t].first], in[que[t].second]};
      return get_block(ps) < get_block(pt);
    };
    sort(all(order), cmp);
    for(int qi: order){
      vi tmp;
      tmp.reserve(n);
      auto move_end = [&](int &a, int b){
        auto step = [&](int c){
          if (on_path[c]){
            remove(a, path); on_path[a] = 0;
          }
          else{
            add(c, path); on_path[c] = 1;
          }
          a = c;
        };
        for(;!(in[b] <= in[a] && in[a] <= out[b]); b =
            pa[b]){
          tmp.push_back(b);
        }
        while (a != b) step(pa[a]);
        for(int i = (int)tmp.size() - 1; i >= 0; i--)
            step(tmp[i]);
        tmp.clear(),tmp.resize(0);
      };
      move_end(path[0], que[qi].first);
      move_end(path[1], que[qi].second);
      ans[qi] = calc(path);
    }
    return ans;
  }
}
```

SegmentTreeLazyIterative.h

Description: Non-recursion segment tree with range updates and range queries.
Usage: T, U are the node and lazy types. TT, UT, UU represents merging two nodes,
applying lazy to a node, and merging two lazies.
Time: $\mathcal{O}\left(\log N\right)$ for each operation
Status: Tested on range add update and range min query.
f55804, 85 lines

```
template<class T,class U> struct segtree{
  //fill merge node + node here
  T TT(T &le, T &ri);

  //fill apply lazy "app" variable to "node" variable
      here
  void UT(U &app, T &node);

  //Append new lazy "node" to existing lazy "app",
      assigning to "app"
  void UU(U &app, U &node);

  //Tid, Uid are default values of T, U,
  //representing empty node and empty lazy (no change).
  T Tid;
  U Uid;
  segtree(T _Tid, U _Uid, vector<T> &a){
    Tid = _Tid, Uid = _Uid;
    build(a);
  }
  vector<T> seg;
  vector<U> d;
  int n, lg;
  void refresh(int v){
    seg[v] = TT(seg[v<<1], seg[(v<<1)|1]);
  }
  //0-index
  void build(vector<T> &a){
    int siz = a.size();
    for(n = 1; n < siz;) n <<= 1;
    seg.assign(n<<1, Tid), d.assign(n, Uid), lg = __lg(n
        );
    for(int i = 0; i < siz; i++) seg[n + i] = a[i];
    for(int i = n - 1; i >= 1; i--) refresh(i);
  }
  void apply(int v, U &f){
    UT(f, seg[v]);
    if(v < n) UU(d[v], f);
  }
  void push(int v){
    apply(v << 1, d[v]), apply((v << 1) | 1, d[v]);
    d[v] = Uid;
  }
  void update(int l, int r, U &f){ //update in [l, r]
    l += n, r += (n + 1);
    if(l >= r) return;
    for(int i = lg; i >= 1; i--){
      if((l >> i << i) != l) push(l >> i);
      if((r >> i << i) != r) push((r - 1) >> i);
    }
    int l0 = l, r0 = r;
```

```
    for(;l < r; l >>= 1, r >>= 1){
      if(l & 1){
        apply(l, f);
        l++;
      }
      if(r & 1){
        r--;
        apply(r, f);
      }
    }
    l = l0, r = r0;
    for(int i = 1; i <= lg; i++){
      if((l >> i << i) != l) refresh(l >> i);
      if((r >> i << i) != r) refresh((r - 1) >> i);
    }
  }
  T query(int l, int r){ // query in [l, r]
    l += n, r += (n + 1);
    if(l >= r) return Tid;
    for(int i = lg; i >= 1; i--){
      if((l >> i << i) != l) push(l >> i);
      if((r >> i << i) != r) push((r - 1) >> i);
    }
    T le = Tid, ri = Tid;
    for(;l < r; l >>= 1, r >>= 1){
      if(l & 1){
        le = TT(le, seg[l]);
        l++;
      }
      if(r & 1){
        r--;
        ri = TT(seg[r], ri);
      }
    }
    return TT(le, ri);
  }
};
```

SplayTree.h

Description: Implicit splay tree with insertion, deletion and access by 1-based sequence position.
Time: $\mathcal{O}(\log N)$ amortized per operation
Status: Tested on SPOJ

dc3105, 116 lines

```
struct Node {
  Node *parent, *child[2];
  int sz = 0, value;

  Node(int val = 0) {
    parent = nullptr, sz = 1, value = val;
    child[0] = child[1] = nullptr;

}
  int side() {
    return parent ? parent->child[1] == this : -1;
  }
  void pushUp() {
    sz = 1;
    for (int i = 0; i < 2; ++i)
      if (child[i]) sz += child[i]->sz;
  }
};
const int NUM_NODES = 1e6 + 5;
Node nodes[NUM_NODES];
int numNode = 0;

Node *newNode(int value = 0) {
  assert(numNode < NUM_NODES);
  nodes[numNode] = Node(value);
  return &nodes[numNode++];
}

struct SplayTree {
  Node *root;

  SplayTree() {
    root = newNode(0);
    link(root, newNode(0), 1);
    root->pushUp();
  }

  Node *oneDir(int dir) {
    Node *p = root;
    while (p && p->child[dir]) p = p->child[dir];
    return p;
  }

  void link(Node *a, Node *b, int dir) {
    if (a == nullptr) {
      root = b;
      if (root) root->parent = nullptr;
      return;
    }
    a->child[dir] = b;
    if (b) b->parent = a;
  }

  void rotate(Node *a) {
    Node *p = a->parent;
    int dir = a->side();
    link(p->parent, a, p->side());
    link(p, a->child[!dir], dir);
    link(a, p, !dir);
    p->pushUp(), a->pushUp();
  }

  void splay(Node *a, Node *s = nullptr) {
    while (a->parent != s) {
      Node *p = a->parent;

      if (p->parent == s) rotate(a);
      else if (a->side() == p->side()) rotate(p), rotate
          (a); // zig-zig
      else rotate(a), rotate(a);
    }
  }

  // Zero-based rank including sentinels; target must be
       below s if s != nullptr.
  void find(int k, Node *s = nullptr) {
    assert(0 <= k && k < root->sz);
    Node *p = root;
    while (true) {
      int cntLeft = (p->child[0] ? p->child[0]->sz : 0);
      if (cntLeft > k) p = p->child[0];
      else if (cntLeft + 1 > k) {
        splay(p, s); return;
      }
      else {
        k -= cntLeft + 1;
        p = p->child[1];
      }
    }
  }

  // Insert before the current k-th element. 1 <= k <= n
       + 1
  void insert(int k, int val) {
    assert(1 <= k && k <= root->sz - 1);
    find(k), find(k - 1, root);
    Node *foo = newNode(val);
    link(foo, root->child[0], 0);
    link(root, foo, 0);
    foo->pushUp(), root->pushUp();
  }

  // 1 <= k <= n
  void remove(int k) {
    assert(1 <= k && k <= root->sz - 2);
    find(k + 1), find(k - 1, root);
    link(root->child[0], nullptr, 1);
    root->child[0]->pushUp();
    root->pushUp();
  }

  // 1 <= k <= n
  int get(int k) {
    assert(1 <= k && k <= root->sz - 2);
    find(k);
    return root->value;
  }
} tree;
```

TimeSegmentTree.h

Description: A time segment tree, used to support deletion in a DS that supports addition and rollback with an additional $\log N$ factor. Note that the queries from the DS has to be offline.

Usage: 1: Edit MAX_TIME, ope, Q (description below)
2: Define add(), rollback() and calc() in namespace segtree_time after including.
3: Turn the addition/deletion updates into tuples [x,l,r],
which means perform update x at time l and delete at time r + 1.
4: Call update(l, r, x) on all tuples, then call solve_queries()
to solve the queries offline.
rollback(k) to rollback k times, unite() and Findset() to merge and find root.
Time: $\mathcal{O}(T(n)\log N)$, where $T(n)$ is the time complexity for the DS.
Status: Tested on one problem

39cb7b, 55 lines

```
namespace segtree_time{ //time is 1-indexed
  //ope is the update type put into the segtree,
  //Q is the type of the query answers.
  using ope = pii;
  using Q = int;
  //MAX_TIME is the maximum time allowed
  const int MAX_TIME = 3e5;
  vector<vector<ope>> group(4 * MAX_TIME + 5);
  void update(int v, int TL, int TR, int l, int r, const
       ope &x){
    if(r < TL || l > TR) return;
    if(l <= TL && TR <= r){
      group[v].push_back(x); return;
    }
    int TM = ((TL + TR) >> 1);
    update(v << 1, TL, TM, l, r, x);
    update((v << 1) | 1, TM + 1, TR, l, r, x);
  }
  void update(int l, int r, const ope &x){
    assert(1 <= l && l <= r && r <= MAX_TIME);
    update(1, 1, MAX_TIME, l, r, x);
  }
  Q calc(int i); // Define to answer the i-th query.
  void add(ope &x); // Define to apply x to your DS.
  //rollback the last k operations of your DS here.
  //Note that ALL operations are counted, EVEN IF they
      don't change the DS.
  void rollback(int k);
  inline void DFS(int v, int TL, int TR, vector<pii> &
      que, int &ptr, vector<Q> &ans){
    if(ptr == que.size()) return;
    for(ope &x: group[v]) add(x);
    if(TL != TR){
      int TM=((TL + TR) >> 1);
      DFS(v << 1, TL, TM, que, ptr, ans);
      DFS((v << 1) | 1,TM + 1, TR, que, ptr, ans);
    }
    else{
      for(; ptr < que.size() && que[ptr].first == TL;
          ptr++){
        ans[que[ptr].second] = calc(que[ptr].second);
      }
    }
    rollback(group[v].size());
  }
  //a query is of format {time_of_query, index_of_query}
  inline vector<Q> solve_queries(vector<pii> &queries){
    for (auto [time, index] : queries) {
      assert(1 <= time && time <= MAX_TIME);
      assert(0 <= index && index < (int)queries.size());
    }
    sort(queries.begin(), queries.end());
    int ptr = 0;
    vector<Q> ans(queries.size());
    DFS(1, 1, MAX_TIME, queries, ptr, ans);
    return ans;
  }
};
using namespace segtree_time;
```

Treap.h
Description: An explicit treap that supports split(), merge(), insert(), and erase().
Usage: Declare pnode t = nullptr for an empty treap,
then use the functions. If you want to handle more data in each node,
then reimplement get_val() and refresh().
Time: $\mathcal{O}(\log N)$
Status: Tested on two problems

f08209, 37 lines

```
mt19937_64 rng(6736);
struct node{
    int x, sz = 1;
    ull y;
    node *le = nullptr, *ri = nullptr;
    node(int x): x(x), y(rng()) {}
};
namespace treap{
    using pnode = node*;
    int get_val(pnode &p){ return p != nullptr ? p->sz :
         0; }
    void refresh(pnode &p){ p->sz = get_val(p->le) +
        get_val(p->ri) + 1; }
    //first is <=x, second is >x
    pair<pnode, pnode> split(pnode t, int x){
        if(t == nullptr) return {nullptr, nullptr};
        pair<pnode, pnode> ans;
        if(t->x <= x){
            ans = split(t->ri, x);
            t->ri = ans.fi; ans.fi = t;
            refresh(ans.fi);
        } else {
            ans = split(t->le, x);
            t->le = ans.se; ans.se = t;
            refresh(ans.se);
        }
        return ans;
    }
    //assume all t1->x < t2->x (ordering between trees)
    pnode merge(pnode t1, pnode t2){
        if(t1 == nullptr || t2 == nullptr) return t1 ==
            nullptr ? t2 : t1;
        if(t1->y >= t2->y){
            t1->ri = merge(t1->ri, t2);
            refresh(t1); return t1;
        }
        t2->le = merge(t1, t2->le);
        refresh(t2); return t2;
    }
}
```

Graph (7)

7.1 Graph theory

7.1.1 Number of Spanning Trees

Create an $N \times N$ matrix mat, and for each edge $a \to b \in G$, do mat[a][b]--, mat[b][b]++ (and mat[b][a]--, mat[a][a]++ if G is undirected). Remove the i-th row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

7.1.2 Erdős–Gallai theorem

A simple graph with node degrees $d_1 \geq \cdots \geq d_n$ exists iff $d_1 + \cdots + d_n$ is even and for every $k = 1 \ldots n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(d_i, k).$$

7.2 Algorithms

TwoSat.h
Description: 2sat -> finding an arbitrary equation
Time: $\mathcal{O}(n+m)$
Status: Tested by KickingKun

485abb, 66 lines

```
struct TwoSat { // convert vector -> array to get higher
     speed ....
  int n; vector <int> lab, num, low; vector <vector <int
      >> adj;
  int timeDfs, scc; stack <int> st;

  TwoSat (int _n) {
    this->n = _n; lab.assign(n * 2 + 1, 0); low.assign(n
         * 2 + 1, 0);
    num.assign(n * 2 + 1, 0); adj.assign(n * 2 + 1,
        vector <int>());
    timeDfs = scc = 0; while (!st.empty()) st.pop();
  }

  int NON(int u) {
    return u <= n ? u + n : u - n;
```

```
  }

  void add(int u, int v) {
    adj[u].emplace_back(v);
  }

  void add_u_or_v(int u, int v) { // u or v = 1
    adj[NON(u)].emplace_back(v); adj[NON(v)].
        emplace_back(u);
  }
  void assign(int u, bool val) { // u = val
    if (val == 1) add_u_or_v(u, u);
    else add_u_or_v(NON(u), NON(u));
  }
  void add_equal(int u, int v) { // u = v
    add_u_or_v(u, NON(v)); add_u_or_v(NON(u), v);
  }
  void add_non_equal(int u, int v) { // u != v
    add_u_or_v(u, v); add_u_or_v(NON(u), NON(v));
  }

  void dfs(int u) {
    low[u] = num[u] = ++timeDfs; st.emplace(u);
    for (int v: adj[u]) {
      if (lab[v]) continue;
      if (!num[v]) dfs(v), low[u] = min(low[u], low[v]);
      else low[u] = min(low[u], num[v]);
    }

    if (low[u] == num[u]) {
      ++scc;
      while (true) {
        int v = st.top(); st.pop();
        lab[v] = scc; if (v == u) break;
      }
    }
  }

  bool check_exist() {
    for (int i = 1; i <= n; i++) {
      if (!lab[NON(i)]) dfs(NON(i));
      if (!lab[i]) dfs(i);
    }
    for (int i = 1; i <= n; i++)
      if (lab[i] == lab[NON(i)]) return false;
    return true;
  }

  vector <int> get_equation() {
    if (!check_exist()) return {-1};
    vector <int> ans = {0};
    for (int i = 1; i <= n; i++) ans.emplace_back(lab[i]
        < lab[NON(i)]);
    return ans;
  }
};
```

BipartieMatching.h

Description: use for max matching

Time: $\mathcal{O}\left(E \cdot \sqrt{V}\right)$

Status: approved by KickingKun

b7e4ae, 82 lines

```
struct bipartite_matching {
  const int n, m;
  vector <int> match1, match2, que, dist;
  vector <vector<int>> adj;

  bipartite_matching(int _n, int _m) : n(_n), m(_m) {
    match1.assign(n + 1, -1); match2.assign(m + 1, -1);
    que.assign(n + 1, 0); dist.assign(n + 1, 0); adj.
        assign(n + 1, vector <int>());
  }

  void add(int u, int v) {
    adj[u].emplace_back(v);
  }

  bool bfs() {
    fill(dist.begin(), dist.end(), -1);
    int queBegin = 1, queEnd = 1;
    for (int u = 1; u <= n; ++u) {
      if (match1[u] == -1) dist[u] = 0, que[queEnd++] =
          u;
    }
    bool success = false;
    while (queBegin < queEnd) {
      int u = que[queBegin++];
      for (int v : adj[u]) {
        if (match2[v] == -1) {
          success = true;
        }
        else if (dist[match2[v]] == -1) {
          dist[match2[v]] = dist[u] + 1;
          que[queEnd++] = match2[v];
        }
      }
    }
    return success;
  }

  bool dfs(int u) {
    for (int v : adj[u]) {
      if (match2[v] == -1 || (dist[match2[v]] == dist[u]
          + 1 && dfs(match2[v]))) {
        match1[u] = v; match2[v] = u; dist[u] = n + m;
        return true;
      }
    }
    dist[u] = n + m;
    return false;
  }

  int max_matching() {
    while (bfs()) {
      for (int u = 1; u <= n; ++u) {
        if (match1[u] == -1) dfs(u);
      }
    }
    return n - count(match1.begin() + 1, match1.end(),
        -1);
  }

  pair <vector <int>, vector <int>> minimum_vertex_cover
      () {
    vector <int> L, R;
    for (int u = 1; u <= n; ++u) {
      if (dist[u] == -1) L.emplace_back(u);
      else if (match1[u] != -1) R.emplace_back(match1[u
          ]);
    }
    return {L, R};
  }

  pair <vector <int>, vector <int>>
      maximum_independent_set() {
    auto [_L, _R] = minimum_vertex_cover();
    vector <int> L, R;
    vector <bool> mark1(n + 2), mark2(m + 2);
    for (int x: _L) mark1[x] = true; for (int x: _R)
        mark2[x] = true;
    for (int u = 1; u <= n; u++) if (!mark1[u]) L.
        emplace_back(u);
    for (int u = 1; u <= m; u++) if (!mark2[u]) R.
        emplace_back(u);
    return {L, R};
  }

  vector <pii> get_edges() { // get from max matching
    vector <pii> ans;
    for (int u = 1; u <= n; ++u)
      if (match1[u] != -1) ans.emplace_back(u, match1[u
          ]);
    return ans;
  }
};
```

MaxFlow.h

Description: Dinitz's algorithm to find max-flow.

Usage: `Call myflow.resize(n) to initalize the network.`

Time: $\mathcal{O}\left(V^2 \cdot E\right)$ in worst cases

Status: Well-tested

3be25f, 72 lines

```
struct Dinic {
  struct Edge {
    int u, v;
    ll capacity, flow = 0;

    Edge(int _u = 0, int _v = 0, ll _capacity = 0) {
      u = _u, v = _v, capacity = _capacity, flow = 0;
    }
```

```
};

const ll FLOW_INF = (ll)2e18;
vector<Edge> edges;
vector<vector<int>> adj;
vector<int> level, ptr;
int n, m;

Dinic(int n = 0) { resize(n); }

void resize(int _n) {
  n = _n, m = 0;
  adj.assign(n + 1, vector<int>()), edges.clear();
}

int add(int u, int v, ll capacity) {
  edges.emplace_back(u, v, capacity);
  edges.emplace_back(v, u, 0);
  adj[u].emplace_back(m++), adj[v].emplace_back(m++);
  return m - 2;
}

ll dfs(int u, int sink, ll pushed) {
  if (u == sink || pushed == 0) return pushed;
  for (int &i = ptr[u]; i < (int)adj[u].size(); ++i) {
    int id = adj[u][i]; Edge e = edges[id];
    if (level[u] + 1 != level[e.v] || e.capacity - e.
        flow <= 0) continue;
    if (ll p = dfs(e.v, sink, min(pushed, e.capacity -
         e.flow))) {
      edges[id].flow += p, edges[id ^ 1].flow -= p;
      return p;
    }
  }
  return 0;
}

ll maxFlow(int source, int sink) {
  assert(source != sink);
  ll flow = 0;
  do {
    level = ptr = vector<int>(n + 1, 0);
    queue<int> q; q.emplace(source); level[source] =
        1;
    while (!q.empty()) {
      int u = q.front(); q.pop();
      for (auto id : adj[u]) {
        if (level[edges[id].v] || edges[id].capacity -
             edges[id].flow <= 0) continue;
        level[edges[id].v] = level[u] + 1;
        q.emplace(edges[id].v);
      }
    }
    while (ll pushed = dfs(source, sink, FLOW_INF))
        flow += pushed;
  } while (level[sink] != 0);
  return flow;
}

vector<pii> minCut() { // remember to call maxFlow(
    source, sink) and be careful of duplicates
  vector<pii> cut;
  for (int id = 0; id < m; id += 2) {
    auto &e = edges[id];
    if (e.capacity > 0 && level[e.u] && !level[e.v])
      cut.emplace_back(e.u, e.v);
  }
  return cut;
}
} myflow;
```

MinCostMaxFlow.h

Description: Edmond-Karp algorithm with SPFA to find the shortest path.
Usage: `Call myflow.resize(n) to initalize the network.`
Time: $\mathcal{O}(F \cdot V \cdot E)$ in worst cases
Status: Well-tested

357dcc, 74 lines

```
struct MCMF {
  struct Edge {
    int u, v;
    ll capacity, cost, flow = 0;

    Edge(int _u = 0, int _v = 0, ll _capacity = 0, ll
        _cost = 0) {
      u = _u, v = _v, capacity = _capacity, cost = _cost
          , flow = 0;
    }
  };

  const ll FLOW_INF = (ll)2e18;
  vector<Edge> edges;
  vector<vector<int>> adj;
  vector<int> level, ptr;
  int n, m;

  MCMF(int n = 0) { resize(n); }

  void resize(int _n) {
    n = _n, m = 0;
    adj.assign(n + 1, vector<int>()), edges.clear();
  }

  int add(int u, int v, ll capacity, ll cost) {
    edges.emplace_back(u, v, capacity, cost);
    edges.emplace_back(v, u, 0, -cost);
    adj[u].emplace_back(m++), adj[v].emplace_back(m++);
    return m - 2;
  }

  bool spfa(int source, int sink, vector<ll> &dist,
      vector<int> &par) {
    dist.assign(n + 1, FLOW_INF), par.assign(n + 1, -1);
    vector<bool> inQueue(n + 1, false); queue<int> q;
    dist[source] = 0; inQueue[source] = true; q.emplace(
        source);
    while (!q.empty()) {
      int u = q.front(); q.pop();
      inQueue[u] = false;
      for (auto id : adj[u]) {
        int v = edges[id].v;
        if (edges[id].capacity - edges[id].flow <= 0)
            continue;
        if (dist[v] > dist[u] + edges[id].cost) {
          dist[v] = dist[u] + edges[id].cost;
          par[v] = id;
          if (!inQueue[v]) {
            inQueue[v] = true;
            q.emplace(v);
          }
        }
      }
    }
    return dist[sink] != FLOW_INF;
  }

  pair<ll, ll> minCost(int source, int sink) {
    assert(source != sink);
    ll flow = 0, cost = 0; vector<ll> dist; vector<int>
        par;
    while (spfa(source, sink, dist, par)) {
      ll amount = FLOW_INF - flow; int cur = sink;
      while (cur != source) {
        amount = min(amount, edges[par[cur]].capacity -
            edges[par[cur]].flow);
        cur = edges[par[cur]].u;
      }

      flow += amount, cost += amount * dist[sink], cur =
          sink;
      while (cur != source) {
        edges[par[cur]].flow += amount;
        edges[par[cur] ^ 1].flow -= amount;
        cur = edges[par[cur]].u;
      }
    }

    return make_pair(flow, cost);
  }
} myflow;
```

BiconnectedComponents.h

Description: Finds all biconnected components in an undirected graph, and runs a callback for the edges in each. In a biconnected component there are at least two internally disjoint paths between any two nodes (a cycle exists through them). Note that a node can be in several components. An edge which is not in a component is a bridge, i.e., not part of any cycle.

Usage: `int eid = 0; ed.resize(N);`
`for each edge (a,b) {`
`ed[a].emplace_back(b, eid);`
`ed[b].emplace_back(a, eid++); }`
`bicomps([&](const vi& edgelist) {...});`
Time: $\mathcal{O}(E+V)$
Status: tested during MIPT ICPC Workshop 2017

c6b7c7, 32 lines

```
vi num, st;
vector<vector<pii>> ed;
int Time;
template<class F>
int dfs(int at, int par, F& f) {
  int me = num[at] = ++Time, top = me;
  for (auto [y, e] : ed[at]) if (e != par) {
    if (num[y]) {
      top = min(top, num[y]);
      if (num[y] < me)
        st.push_back(e);
    } else {
      int si = sz(st);
      int up = dfs(y, e, f);
      top = min(top, up);
      if (up == me) {
        st.push_back(e);
        f(vi(st.begin() + si, st.end()));
        st.resize(si);
      }
      else if (up < me) st.push_back(e);
      else { /* e is a bridge */ }
    }
  }
  return top;
}

template<class F>
void bicomps(F f) {
  num.assign(sz(ed), 0);
  rep(i,0,sz(ed)) if (!num[i]) dfs(i, -1, f);
}
```

DFSMatching.h

Description: Simple bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and *btoa* should be a vector full of -1's of the same size as the right partition. Returns the size of the matching. *btoa*[i] will be the match for vertex i on the right side, or -1 if it's not matched.
Usage: `vi btoa(m, -1); dfsMatching(g, btoa);`
Time: $\mathcal{O}(VE)$
Status: works

522b98, 22 lines

```
bool find(int j, vector<vi>& g, vi& btoa, vi& vis) {
  if (btoa[j] == -1) return 1;
  vis[j] = 1; int di = btoa[j];
  for (int e : g[di])
    if (!vis[e] && find(e, g, btoa, vis)) {
      btoa[e] = di;
      return 1;
    }
  return 0;
}
int dfsMatching(vector<vi>& g, vi& btoa) {
  vi vis;
  rep(i,0,sz(g)) {
    vis.assign(sz(btoa), 0);
    for (int j : g[i])
      if (find(j, g, btoa, vis)) {
        btoa[j] = i;
        break;
      }
  }
  return sz(btoa) - (int)count(all(btoa), -1);
}
```

DirectedMST.h

Description: Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.
Time: $\mathcal{O}(E \log V)$
Status: Stress-tested, also tested on NWERC 2018 fastestspeedrun

"../data-structures/UnionFindRollback.h" 39e620, 60 lines

```
struct Edge { int a, b; ll w; };
struct Node {
  Edge key;
  Node *l, *r;
  ll delta;
  void prop() {
    key.w += delta;
    if (l) l->delta += delta;
    if (r) r->delta += delta;
    delta = 0;
  }
  Edge top() { prop(); return key; }
};
Node *merge(Node *a, Node *b) {
  if (!a || !b) return a ?: b;
  a->prop(), b->prop();
  if (a->key.w > b->key.w) swap(a, b);
  swap(a->l, (a->r = merge(b, a->r)));
  return a;
}
void pop(Node*& a) { a->prop(); a = merge(a->l, a->r); }

pair<ll, vi> dmst(int n, int r, vector<Edge>& g) {
  RollbackUF uf(n);
  vector<Node*> heap(n);
  for (Edge e : g) heap[e.b] = merge(heap[e.b], new Node
      {e});
  ll res = 0;
  vi seen(n, -1), path(n), par(n);
  seen[r] = r;
  vector<Edge> Q(n), in(n, {-1,-1}), comp;
  deque<tuple<int, int, vector<Edge>>> cycs;
  rep(s,0,n) {
    int u = s, qi = 0, w;
    while (seen[u] < 0) {
      if (!heap[u]) return {-1,{}};
      Edge e = heap[u]->top();
      heap[u]->delta -= e.w, pop(heap[u]);
      Q[qi] = e, path[qi++] = u, seen[u] = s;
      res += e.w, u = uf.find(e.a);
      if (seen[u] == s) {
        Node* cyc = 0;
        int end = qi, time = uf.time();
        do cyc = merge(cyc, heap[w = path[--qi]]);
        while (uf.join(u, w));
        u = uf.find(u), heap[u] = cyc, seen[u] = -1;
        cycs.push_front({u, time, {&Q[qi], &Q[end]}});
      }
    }
    rep(i,0,qi) in[uf.find(Q[i].b)] = Q[i];
  }

  for (auto& [u,t,comp] : cycs) { // restore sol (
      optional)
    uf.rollback(t);
    Edge inEdge = in[u];
    for (auto& e : comp) in[uf.find(e.b)] = e;
    in[uf.find(inEdge.b)] = inEdge;
  }
  rep(i,0,n) par[i] = in[i].a;
  return {res, par};
}
```

EdgeColoring.h

Description: Given a simple, undirected graph with max degree D, computes a $(D+1)$-coloring of the edges such that no neighboring edges share a color. (D-coloring is NP-hard, but can be done for bipartite graphs by repeated matchings of max-degree nodes.)
Time: $\mathcal{O}(NM)$
Status: stress-tested, tested on kattis:gamescheduling

e210e2, 31 lines

```
vi edgeColoring(int N, vector<pii> eds) {
  vi cc(N + 1), ret(sz(eds)), fan(N), free(N), loc;
  for (pii e : eds) ++cc[e.first], ++cc[e.second];
  int u, v, ncols = *max_element(all(cc)) + 1;
  vector<vi> adj(N, vi(ncols, -1));
  for (pii e : eds) {
    tie(u, v) = e;
    fan[0] = v;
    loc.assign(ncols, 0);
    int at = u, end = u, d, c = free[u], ind = 0, i = 0;
    while (d = free[v], !loc[d] && (v = adj[u][d]) !=
        -1)
      loc[d] = ++ind, cc[ind] = d, fan[ind] = v;
    cc[loc[d]] = c;
    for (int cd = d; at != -1; cd ^= c ^ d, at = adj[at
        ][cd])
      swap(adj[at][cd], adj[end = at][cd ^ c ^ d]);
    while (adj[fan[i]][d] != -1) {
      int left = fan[i], right = fan[++i], e = cc[i];
```

```
      adj[u][e] = left;
      adj[left][e] = u;
      adj[right][e] = -1;
      free[right] = e;
    }
    adj[u][d] = fan[i];
    adj[fan[i]][d] = u;
    for (int y : {fan[0], u, end})
      for (int& z = free[y] = 0; adj[y][z] != -1; z++);
  }
  rep(i,0,sz(eds))
    for (tie(u, v) = eds[i]; adj[u][ret[i]] != v;) ++ret
        [i];
  return ret;
}
```

EulerWalk.h

Description: Eulerian undirected/directed path/cycle algorithm. Input should be a vector of (dest, global edge index), where for undirected graphs, forward/backward edges have the same index. Returns a list of nodes in the Eulerian path/cycle with src at both start and end, or empty list if no cycle/path exists. To get edge indices back, add .second to s and ret.

Time: $\mathcal{O}(V+E)$

Status: stress-tested

780b64, 15 lines

```
vi eulerWalk(vector<vector<pii>>& gr, int nedges, int
    src=0) {
  int n = sz(gr);
  vi D(n), its(n), eu(nedges), ret, s = {src};
  D[src]++; // to allow Euler paths, not just cycles
  while (!s.empty()) {
    int x = s.back(), y, e, &it = its[x], end = sz(gr[x
        ]);
    if (it == end){ ret.push_back(x); s.pop_back();
        continue; }
    tie(y, e) = gr[x][it++];
    if (!eu[e]) {
      D[x]--, D[y]++;
      eu[e] = 1; s.push_back(y);
    }}
  for (int x : D) if (x < 0 || sz(ret) != nedges+1)
      return {};
  return {ret.rbegin(), ret.rend()};
}
```

LinkCutTree.h

Description: Represents a forest of unrooted trees. You can add and remove edges (as long as the result is still a forest), and check whether two nodes are in the same tree.

Time: All operations take amortized $\mathcal{O}(\log N)$.

Status: Stress-tested a bit for N <= 20

0fb462, 90 lines

```
struct Node { // Splay tree. Root's pp contains tree's
    parent.
  Node *p = 0, *pp = 0, *c[2];
  bool flip = 0;
  Node() { c[0] = c[1] = 0; fix(); }
  void fix() {
    if (c[0]) c[0]->p = this;
    if (c[1]) c[1]->p = this;
    // (+ update sum of subtree elements etc. if wanted)
  }
  void pushFlip() {
    if (!flip) return;
    flip = 0; swap(c[0], c[1]);
    if (c[0]) c[0]->flip ^= 1;
    if (c[1]) c[1]->flip ^= 1;
  }
  int up() { return p ? p->c[1] == this : -1; }
  void rot(int i, int b) {
    int h = i ^ b;
    Node *x = c[i], *y = b == 2 ? x : x->c[h], *z = b ?
        y : x;
    if ((y->p = p)) p->c[up()] = y;
    c[i] = z->c[i ^ 1];
    if (b < 2) {
      x->c[h] = y->c[h ^ 1];
      y->c[h ^ 1] = x;
    }
    z->c[i ^ 1] = this;
    fix(); x->fix(); y->fix();
    if (p) p->fix();
    swap(pp, y->pp);
  }
  void splay() {
    for (pushFlip(); p; ) {
      if (p->p) p->p->pushFlip();
      p->pushFlip(); pushFlip();
      int c1 = up(), c2 = p->up();
      if (c2 == -1) p->rot(c1, 2);
      else p->p->rot(c2, c1 != c2);
    }
  }
  Node* first() {
    pushFlip();
    return c[0] ? c[0]->first() : (splay(), this);
  }
};

struct LinkCut {
  vector<Node> node;
  LinkCut(int N) : node(N) {}

  void link(int u, int v) { // add an edge (u, v)
    assert(!connected(u, v));
    makeRoot(&node[u]);
    node[u].pp = &node[v];
  }
  void cut(int u, int v) { // remove an edge (u, v)
    Node *x = &node[u], *top = &node[v];
    makeRoot(top); x->splay();
    assert(top == (x->pp ?: x->c[0]));
    if (x->pp) x->pp = 0;
    else {
      x->c[0] = top->p = 0;
      x->fix();
    }
  }
  bool connected(int u, int v) { // are u, v in the same
      tree?
    Node* nu = access(&node[u])->first();
    return nu == access(&node[v])->first();
  }
  void makeRoot(Node* u) {
    access(u);
    u->splay();
    if(u->c[0]) {
      u->c[0]->p = 0;
      u->c[0]->flip ^= 1;
      u->c[0]->pp = u;
      u->c[0] = 0;
      u->fix();
    }
  }
  Node* access(Node* u) {
    u->splay();
    while (Node* pp = u->pp) {
      pp->splay(); u->pp = 0;
      if (pp->c[1]) {
        pp->c[1]->p = 0; pp->c[1]->pp = pp; }
      pp->c[1] = u; pp->fix(); u = pp;
    }
    return u;
  }
};
```

PushRelabel.h

Description: Push-relabel using the highest label selection rule and the gap heuristic. Quite fast in practice. To obtain the actual flow, look at positive values only.

Time: $\mathcal{O}\left(V^2\sqrt{E}\right)$

Status: Tested on Kattis and SPOJ, and stress-tested

0ae1d4, 48 lines

```
struct PushRelabel {
  struct Edge {
    int dest, back;
    ll f, c;
  };
  vector<vector<Edge>> g;
  vector<ll> ec;
  vector<Edge*> cur;
  vector<vi> hs; vi H;
  PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n
      ) {}

  void addEdge(int s, int t, ll cap, ll rcap=0) {
    if (s == t) return;
    g[s].push_back({t, sz(g[t]), 0, cap});
```

```
    g[t].push_back({s, sz(g[s])-1, 0, rcap});
  }

  void addFlow(Edge& e, ll f) {
    Edge &back = g[e.dest][e.back];
    if (!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest
        );
    e.f += f; e.c -= f; ec[e.dest] += f;
    back.f -= f; back.c += f; ec[back.dest] -= f;
  }
  ll calc(int s, int t) {
    int v = sz(g); H[s] = v; ec[t] = 1;
    vi co(2*v); co[0] = v-1;
    rep(i,0,v) cur[i] = g[i].data();
    for (Edge& e : g[s]) addFlow(e, e.c);

    for (int hi = 0;;) {
      while (hs[hi].empty()) if (!hi--) return -ec[s];
      int u = hs[hi].back(); hs[hi].pop_back();
      while (ec[u] > 0)  // discharge u
        if (cur[u] == g[u].data() + sz(g[u])) {
          H[u] = 1e9;
          for (Edge& e : g[u]) if (e.c && H[u] > H[e.
              dest]+1)
            H[u] = H[e.dest]+1, cur[u] = &e;
          if (++co[H[u]], !--co[hi] && hi < v)
            rep(i,0,v) if (hi < H[i] && H[i] < v)
              --co[H[i]], H[i] = v + 1;
          hi = H[u];
        } else if (cur[u]->c && H[u] == H[cur[u]->dest
            ]+1)
          addFlow(*cur[u], min(ec[u], cur[u]->c));
        else ++cur[u];
    }
  }
  bool leftOfMinCut(int a) { return H[a] >= sz(g); }
};
```

SCC.h

Description: Finds strongly connected components in a directed graph. If vertices u, v belong to the same component, we can reach u from v and vice versa.
Usage: scc(graph, [&](vi& v) { ... }) visits all components in reverse topological order. comp[i] holds the component index of a node (a component only has edges to components with lower index). ncomps will contain the number of components.
Time: $\mathcal{O}(E+V)$
Status: Bruteforce-tested for N <= 5

76b5c9, 24 lines

```
vi val, comp, z, cont;
int Time, ncomps;
template<class G, class F> int dfs(int j, G& g, F& f) {
  int low = val[j] = ++Time, x; z.push_back(j);
  for (auto e : g[j]) if (comp[e] < 0)
    low = min(low, val[e] ?: dfs(e,g,f));

  if (low == val[j]) {
    do {
      x = z.back(); z.pop_back();
      comp[x] = ncomps;
      cont.push_back(x);
    } while (x != j);
    f(cont); cont.clear();
    ncomps++;
  }
  return val[j] = low;
}
template<class G, class F> void scc(G& g, F f) {
  int n = sz(g);
  val.assign(n, 0); comp.assign(n, -1);
  Time = ncomps = 0;
  rep(i,0,n) if (comp[i] < 0) dfs(i, g, f);
}
```

WeightedMatching.h

Description: Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires $N \leq M$.
Time: $\mathcal{O}\left(N^2M\right)$
Status: Tested on kattis:cordonbleu, stress-tested

1e0fe9, 31 lines

```
pair<int, vi> hungarian(const vector<vi> &a) {
  if (a.empty()) return {0, {}};
  int n = sz(a) + 1, m = sz(a[0]) + 1;
  vi u(n), v(m), p(m), ans(n - 1);
  rep(i,1,n) {
    p[0] = i;
    int j0 = 0; // add "dummy" worker 0
    vi dist(m, INT_MAX), pre(m, -1);
    vector<bool> done(m + 1);
    do { // dijkstra
      done[j0] = true;
      int i0 = p[j0], j1, delta = INT_MAX;
      rep(j,1,m) if (!done[j]) {
        auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
        if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
        if (dist[j] < delta) delta = dist[j], j1 = j;
      }
      rep(j,0,m) {
        if (done[j]) u[p[j]] += delta, v[j] -= delta;
        else dist[j] -= delta;
      }
      j0 = j1;
    } while (p[j0]);
    while (j0) { // update alternating path
      int j1 = pre[j0];
      p[j0] = p[j1], j0 = j1;
    }
  }
  rep(j,1,m) if (p[j]) ans[p[j] - 1] = j - 1;
  return {-v[0], ans}; // min cost
}
```

Geometry (8)

Geometry.h

Description: Geometric Template
Time: $\mathcal{O}(1)$
Status: Seems OK

2dde4c, 136 lines

```
struct Point {
  double x, y;
  Point() {}
  Point (double _x, double _y) {
    this->x = _x; this->y = _y;
  }

  double len() {
    return x * x + y * y;
  }

  Point operator - (Point other) {
    return {x - other.x, y - other.y};
  }

  Point operator - () {
    return Point(-x, -y);
  }

  bool operator == (Point other) {
    return x == other.x && y == other.y;
  }

  bool operator != (Point other) {
    return x != other.x || y != other.y;
  }

  void show() {
    cout << x << ' ' << y << '\n';
  }
};

struct Line {
  double a, b, c;
  Line() {}
  Line (double _a, double _b, double _c) {
    this->a = _a; this->b = _b; this->c = _c;
  }

  void simplify() { // if double = ll
    double d = __gcd(__gcd((ll)a, (ll)b), (ll)c);
    a /= d, b /= d, c /= d;
    // else do nothing
```

```
  }
  void show() {
    cout << a << ' ' << b << ' ' << c << '\n';
  }
};

double dot(Point A, Point B) { // dot product
  return A.x * B.x + A.y * B.y;
}

double cross(Point A, Point B) { // cross product
  return A.x * B.y - A.y * B.x;
}

int sign(double x) {
  if (x < 0) return -1;
  if (x == 0) return 0;
  return 1;
}

double dis(Point A, Point B) {
  return sqrt((B.x - A.x) * (B.x - A.x) + (B.y - A.y) *
      (B.y - A.y));
}

double dis(Point m, Line d) { // d(M, delta)
  return abs(d.a * m.x + d.b * m.y + d.c) / sqrt(d.a * d
      .a + d.b * d.b);
}

double polygon_area(vector <Point> p) {
  if (p.size() < 3) return 0;
  double res = 0;
  for (int i = 1; i < p.size() - 1; i++)
    res += cross(p[i] - p[0], p[i + 1] - p[0]);
  return abs(res) / 2.0;
} // tested

bool same_side(Point A, Point B, Line d) { // out of
    line
  double y1 = d.a * A.x + d.b * A.y + d.c;
  double y2 = d.a * B.x + d.b * B.y + d.c;
  return (y1 > 0 && y2 > 0) || (y1 < 0 && y2 < 0);
} // tested by eyes

Line lineVector(Point A, Point n) { // Point + vector n
  return Line(n.x, n.y, -n.x * A.x - n.y * A.y);
}

Line line(Point A, Point B) { // Point A, B
  Point n(A.y - B.y, B.x - A.x);
  return Line(n.x, n.y, -n.x * A.x - n.y * A.y);
}

bool inside_line(Point A, Line d) { // inside of line
  return d.a * A.x + d.b * A.y + d.c == 0;
}

bool inside_segment(Point M, Point A, Point B) { //
    inside of segment AB
  if (M.x > max(A.x, B.x) || M.x < min(A.x, B.x)) return
      false;
  if (M.y > max(A.y, B.y) || M.y < min(A.y, B.y)) return
      false;
  return inside_line(M, line(A, B));
} // tested CSES / VNOI

Point intersect(Line A, Line B) { // find p(x, y)
  Point res; if (A.b == 0) swap(A, B);
  res.x = (B.c * A.b - A.c * B.b) / (A.a * B.b - B.a * A
      .b);
  res.y = (-A.c - A.a * res.x) / A.b;
  return res;
} // seem OK

Line perpendicular(Point A, Line d) {
  Point n(-d.b, d.a);
  return Line(n.x, n.y, -n.x * A.x - n.y * A.y);
} // tested

bool obtuse(Point M, Point A, Point B) { // MAB > 90
  return (M - A).len() + (B - A).len() < (M - B).len();
} // tested

double dist(Point M, Point A, Point B) { // dis(M,
    segment AB)
if (A == B) return dis(M, A);
//  check MBA <= 90 && MAB <= 90
  if (obtuse(M, A, B) || obtuse(M, B, A)) return min(dis
      (M, A), dis(M, B));
  return dis(M, line(A, B));
} // tested at VNOI

bool intersect(Point A, Point B, Point C, Point D) { //
    segment AB and CD
  int x = sign(cross(C - A, B - A)), y = sign(cross(D -
      A, B - A));
  int u = sign(cross(A - C, D - C)), v = sign(cross(B -
      C, D - C));
  if (x == 0 && inside_segment(C, A, B)) return true;
  if (y == 0 && inside_segment(D, A, B)) return true;
  if (u == 0 && inside_segment(A, C, D)) return true;
  if (v == 0 && inside_segment(B, C, D)) return true;
  return (x != y && u != v);
} // tested at CSES
```

3dHull.h

Description: Computes all faces of the 3-dimension hull of a point set. *No four points must be coplanar*, or else random results will be returned. All faces will point outwards.

Time: $\mathcal{O}\left(n^2\right)$

Status: tested on SPOJ CH3D

"Point3D.h" 5b45fc, 49 lines

```
typedef Point3D<double> P3;

struct PR {
  void ins(int x) { (a == -1 ? a : b) = x; }
  void rem(int x) { (a == x ? a : b) = -1; }
  int cnt() { return (a != -1) + (b != -1); }
  int a, b;
};

struct F { P3 q; int a, b, c; };

vector<F> hull3d(const vector<P3>& A) {
  assert(sz(A) >= 4);
  vector<vector<PR>> E(sz(A), vector<PR>(sz(A), {-1, -1}
      ));
#define E(x,y) E[f.x][f.y]
  vector<F> FS;
  auto mf = [&](int i, int j, int k, int l) {
    P3 q = (A[j] - A[i]).cross((A[k] - A[i]));
    if (q.dot(A[l]) > q.dot(A[i]))
      q = q * -1;
    F f{q, i, j, k};
    E(a,b).ins(k); E(a,c).ins(j); E(b,c).ins(i);
    FS.push_back(f);
  };
  rep(i,0,4) rep(j,i+1,4) rep(k,j+1,4)
    mf(i, j, k, 6 - i - j - k);

  rep(i,4,sz(A)) {
    rep(j,0,sz(FS)) {
      F f = FS[j];
      if(f.q.dot(A[i]) > f.q.dot(A[f.a])) {
        E(a,b).rem(f.c);
        E(a,c).rem(f.b);
        E(b,c).rem(f.a);
        swap(FS[j--], FS.back());
        FS.pop_back();
      }
    }
    int nw = sz(FS);
    rep(j,0,nw) {
      F f = FS[j];
#define C(a, b, c) if (E(a,b).cnt() != 2) mf(f.a, f.b, i
    , f.c);
      C(a, b, c); C(a, c, b); C(b, c, a);
    }
  }
  for (F& it : FS) if ((A[it.b] - A[it.a]).cross(
    A[it.c] - A[it.a]).dot(it.q) <= 0) swap(it.c, it.b);
  return FS;
};
```

Angle.h

Description: A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.
Usage: `vector<Angle> v = {w[0], w[0].t360() ...}; // sorted`
`int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }`
`// sweeps j such that (j-i) represents the number of positively oriented triangles with vertices at 0 and i`
Status: Used, works well

0f0602, 35 lines

```
struct Angle {
  int x, y;
  int t;
  Angle(int x, int y, int t=0) : x(x), y(y), t(t) {}
  Angle operator-(Angle b) const { return {x-b.x, y-b.y,
       t}; }
  int half() const {
    assert(x || y);
    return y < 0 || (y == 0 && x < 0);
  }
  Angle t90() const { return {-y, x, t + (half() && x >=
       0)}; }
  Angle t180() const { return {-x, -y, t + half()}; }
  Angle t360() const { return {x, y, t + 1}; }
};
bool operator<(Angle a, Angle b) {
  // add a.dist2() and b.dist2() to also compare
       distances
  return make_tuple(a.t, a.half(), a.y * (ll)b.x) <
       make_tuple(b.t, b.half(), a.x * (ll)b.y);
}

// Given two points, this calculates the smallest angle
     between
// them, i.e., the angle that covers the defined line
     segment.
pair<Angle, Angle> segmentAngles(Angle a, Angle b) {
  if (b < a) swap(a, b);
  return (b < a.t180() ?
      make_pair(a, b) : make_pair(b, a.t360()));
}
Angle operator+(Angle a, Angle b) { // point a + vector
     b
  Angle r(a.x + b.x, a.y + b.y, a.t);
  if (a.t180() < r) r.t--;
  return r.t180() < a ? r.t360() : r;
}
Angle angleDiff(Angle a, Angle b) { // angle b - angle a
  int tu = b.t - a.t; a.t = b.t;
  return {a.x*b.x + a.y*b.y, a.x*b.y - a.y*b.x, tu - (b
       < a)};
}
```

CircleIntersection.h

Description: Computes the pair of points at which two circles intersect. Returns false in case of no intersection.
Status: stress-tested

"Point.h" 84d6d3, 11 lines

```
typedef Point<double> P;
bool circleInter(P a,P b,double r1,double r2,pair<P, P>*
     out) {
  if (a == b) { assert(r1 != r2); return false; }
  P vec = b - a;
  double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
       p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p
           *d2;
  if (sum*sum < d2 || dif*dif > d2) return false;
  P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2)
       / d2);
  *out = {mid + per, mid - per};
  return true;
}
```

CircleLine.h

Description: Finds the intersection between a circle and a line. Returns a vector of either 0, 1, or 2 intersection points. P is intended to be Point<double>.
Status: unit tested

"Point.h" e0cfba, 9 lines

```
template<class P>
vector<P> circleLine(P c, double r, P a, P b) {
  P ab = b - a, p = a + ab * (c-a).dot(ab) / ab.dist2();
  double s = a.cross(b, c), h2 = r*r - s*s / ab.dist2();
  if (h2 < 0) return {};
  if (h2 == 0) return {p};
  P h = ab.unit() * sqrt(h2);
  return {p - h, p + h};
}
```

CirclePolygonIntersection.h

Description: Returns the area of the intersection of a circle with a ccw polygon.
Time: $\mathcal{O}(n)$
Status: Tested on GNYR 2019 Gerrymandering, stress-tested

"../../content/geometry/Point.h" 19add1, 19 lines

```
typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
  auto tri = [&](P p, P q) {
    auto r2 = r * r / 2;
    P d = q - p;
    auto a = d.dot(p)/d.dist2(), b = (p.dist2()-r*r)/d.
        dist2();
    auto det = a * a - b;
    if (det <= 0) return arg(p, q) * r2;
    auto s = max(0., -a-sqrt(det)), t = min(1., -a+sqrt(
        det));
    if (t < 0 || 1 <= s) return arg(p, q) * r2;
    P u = p + d * s, v = q + d * (t-1);
    return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) * r2;
  };
  auto sum = 0.0;
  rep(i,0,sz(ps))
    sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
  return sum;
}
```

CircleTangents.h

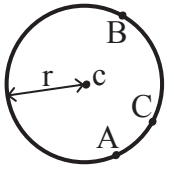
Description: Finds the external tangents of two circles, or internal if r2 is negated. Can return 0, 1, or 2 tangents – 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r2 to 0.
Status: tested

"Point.h" b0153d, 13 lines

```
template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2,
     double r2) {
  P d = c2 - c1;
  double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr
       ;
  if (d2 == 0 || h2 < 0)  return {};
  vector<pair<P, P>> out;
  for (double sign : {-1, 1}) {
    P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
    out.push_back({c1 + v * r1, c2 + v * r2});
  }
  if (h2 == 0) out.pop_back();
  return out;
}
```

circumcircle.h

Description: The circumcirle of a triangle is the circle intersecting all three vertices. ccRadius returns the radius of the circle going through points A, B and C and ccCenter returns the center of the same circle.



Status: tested

"Point.h" 1caa3a, 9 lines

```
typedef Point<double> P;
double ccRadius(const P& A, const P& B, const P& C) {
  return (B-A).dist()*(C-B).dist()*(A-C).dist()/
      abs((B-A).cross(C-A))/2;
}
P ccCenter(const P& A, const P& B, const P& C) {
  P b = C-A, c = B-A;
```

```
  return A + (b*c.dist2()-c*b.dist2()).perp()/b.cross(c)
      /2;
}
```

ClosestPair.h

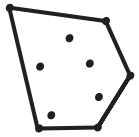
Description: Finds the closest pair of points.
Time: $\mathcal{O}(n \log n)$
Status: stress-tested

"Point.h" ac41a6, 17 lines

```
typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
  assert(sz(v) > 1);
  set<P> S;
  sort(all(v), [](P a, P b) { return a.y < b.y; });
  pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
  int j = 0;
  for (P p : v) {
    P d{1 + (ll)sqrt(ret.first), 0};
    while (v[j].y <= p.y - d.x) S.erase(v[j++]);
    auto lo = S.lower_bound(p - d), hi = S.upper_bound(p
        + d);
    for (; lo != hi; ++lo)
      ret = min(ret, {(*lo - p).dist2(), {*lo, p}});
    S.insert(p);
  }
  return ret.second;
}
```

ConvexHull.h

Description: Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull.



Time: $\mathcal{O}(n \log n)$
Status: stress-tested, tested with kattis:convexhull

"Point.h" 310954, 13 lines

```
typedef Point<ll> P;
vector<P> convexHull(vector<P> pts) {
  if (sz(pts) <= 1) return pts;
  sort(all(pts));
  vector<P> h(sz(pts)+1);
  int s = 0, t = 0;
  for (int it = 2; it--; s = --t, reverse(all(pts)))
    for (P p : pts) {
      while (t >= s + 2 && h[t-2].cross(h[t-1], p) <= 0)
          t--;
      h[t++] = p;
    }
  return {h.begin(), h.begin() + t - (t == 2 && h[0] ==
      h[1])};
}
```

DelaunayTriangulation.h

Description: Computes the Delaunay triangulation of a set of points. Each circumcircle contains none of the input points. If any three points are collinear or any four are on the same circle, behavior is undefined.
Time: $\mathcal{O}(n^2)$
Status: stress-tested

"Point.h", "3dHull.h" c0e7bc, 10 lines

```
template<class P, class F>
void delaunay(vector<P>& ps, F trifun) {
  if (sz(ps) == 3) { int d = (ps[0].cross(ps[1], ps[2])
      < 0);
    trifun(0,1+d,2-d); }
  vector<P3> p3;
  for (P p : ps) p3.emplace_back(p.x, p.y, p.dist2());
  if (sz(ps) > 3) for(auto t:hull3d(p3)) if ((p3[t.b]-p3
      [t.a]).
      cross(p3[t.c]-p3[t.a]).dot(P3(0,0,1)) < 0)
    trifun(t.a, t.c, t.b);
}
```

FastDelaunay.h

Description: Fast Delaunay triangulation. Each circumcircle contains none of the input points. There must be no duplicate points. If all points are on a line, no triangles will be returned. Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in order {t[0][0], t[0][1], t[0][2], t[1][0], ... }, all counter-clockwise.
Time: $\mathcal{O}(n \log n)$
Status: stress-tested

"Point.h" eefdf5, 88 lines

```
typedef Point<ll> P;
typedef struct Quad* Q;
typedef __int128_t lll; // (can be ll if coords are < 2
    e4)
P arb(LLONG_MAX,LLONG_MAX); // not equal to any other
    point

struct Quad {
  Q rot, o; P p = arb; bool mark;
  P& F() { return r()->p; }
  Q& r() { return rot->rot; }
  Q prev() { return rot->o->rot; }
  Q next() { return r()->prev(); }
} *H;

bool circ(P p, P a, P b, P c) { // is p in the
    circumcircle?
  lll p2 = p.dist2(), A = a.dist2()-p2,
    B = b.dist2()-p2, C = c.dist2()-p2;
  return p.cross(a,b)*C + p.cross(b,c)*A + p.cross(c,a)*
      B > 0;
}
Q makeEdge(P orig, P dest) {
  Q r = H ? H : new Quad{new Quad{new Quad{new Quad{0}}}
      };
  H = r->o; r->r()->r() = r;
  rep(i,0,4) r = r->rot, r->p = arb, r->o = i & 1 ? r :
      r->r();
  r->p = orig; r->F() = dest;
  return r;
}
void splice(Q a, Q b) {
  swap(a->o->rot->o, b->o->rot->o); swap(a->o, b->o);
}
Q connect(Q a, Q b) {
  Q q = makeEdge(a->F(), b->p);
  splice(q, a->next());
  splice(q->r(), b);
  return q;
}

pair<Q,Q> rec(const vector<P>& s) {
  if (sz(s) <= 3) {
    Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.
        back());
    if (sz(s) == 2) return { a, a->r() };
    splice(a->r(), b);
    auto side = s[0].cross(s[1], s[2]);
    Q c = side ? connect(b, a) : 0;
    return {side < 0 ? c->r() : a, side < 0 ? c : b->r()
        };
  }

#define H(e) e->F(), e->p
#define valid(e) (e->F().cross(H(base)) > 0)
  Q A, B, ra, rb;
  int half = sz(s) / 2;
  tie(ra, A) = rec({all(s) - half});
  tie(B, rb) = rec({sz(s) - half + all(s)});
  while ((B->p.cross(H(A)) < 0 && (A = A->next())) ||
      (A->p.cross(H(B)) > 0 && (B = B->r()->o)));
  Q base = connect(B->r(), A);
  if (A->p == ra->p) ra = base->r();
  if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e))
    \
    while (circ(e->dir->F(), H(base), e->F())) { \
      Q t = e->dir; \
      splice(e, e->prev()); \
      splice(e->r(), e->r()->prev()); \
      e->o = H; H = e; e = t; \
    }
  for (;;) {
    DEL(LC, base->r(), o);  DEL(RC, base, prev());
    if (!valid(LC) && !valid(RC)) break;
    if (!valid(LC) || (valid(RC) && circ(H(RC), H(LC))))
      base = connect(RC, base->r());
    else
```

```
      base = connect(base->r(), LC->r());
  }
  return { ra, rb };
}

vector<P> triangulate(vector<P> pts) {
  sort(all(pts));  assert(unique(all(pts)) == pts.end())
      ;
  if (sz(pts) < 2) return {};
  Q e = rec(pts).first;
  vector<Q> q = {e};
  int qi = 0;
  while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.push_back(c
    ->p); \
  q.push_back(c->r()); c = c->next(); } while (c != e);
      }
  ADD; pts.clear();
  while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
  return pts;
}
```

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).
Time: $\mathcal{O}(n)$
Status: stress-tested, tested on kattis:roberthood

"Point.h" c571b8, 12 lines

```
typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
  int n = sz(S), j = n < 2 ? 0 : 1;
  pair<ll, array<P, 2>> res({0, {S[0], S[0]}});
  rep(i,0,j)
    for (;; j = (j + 1) % n) {
      res = max(res, {(S[i] - S[j]).dist2(), {S[i], S[j]
          }});
      if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i])
          >= 0)
        break;
    }
  return res.second;
}
```

InsidePolygon.h

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow.
Usage: `vector<P> v = {P{4,4}, P{1,2}, P{2,1}};`
`bool in = inPolygon(v, P{3, 3}, false);`
Time: $\mathcal{O}(n)$
Status: stress-tested and tested on kattis:pointinpolygon

"Point.h", "OnSegment.h", "SegmentDistance.h" 2bf504, 11 lines

```
template<class P>
bool inPolygon(vector<P> &p, P a, bool strict = true) {
  int cnt = 0, n = sz(p);
  rep(i,0,n) {
    P q = p[(i + 1) % n];
    if (onSegment(p[i], q, a)) return !strict;
    //or: if (segDist(p[i], q, a) <= eps) return !strict
        ;
    cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i], q)
        > 0;
  }
  return cnt;
}
```

kdTree.h

Description: KD-tree (2d, can be extended to 3d)
Status: Tested on excellentengineers

"Point.h" bac5b0, 63 lines

```
typedef long long T;
typedef Point<T> P;
const T INF = numeric_limits<T>::max();

bool on_x(const P& a, const P& b) { return a.x < b.x; }
bool on_y(const P& a, const P& b) { return a.y < b.y; }

struct Node {
  P pt; // if this is a leaf, the single point in it
  T x0 = INF, x1 = -INF, y0 = INF, y1 = -INF; // bounds
  Node *first = 0, *second = 0;

  T distance(const P& p) { // min squared distance to a
      point
    T x = (p.x < x0 ? x0 : p.x > x1 ? x1 : p.x);
    T y = (p.y < y0 ? y0 : p.y > y1 ? y1 : p.y);
    return (P(x,y) - p).dist2();
  }

  Node(vector<P>&& vp) : pt(vp[0]) {
    for (P p : vp) {
      x0 = min(x0, p.x); x1 = max(x1, p.x);
      y0 = min(y0, p.y); y1 = max(y1, p.y);
    }
    if (vp.size() > 1) {
      // split on x if width >= height (not ideal...)
      sort(all(vp), x1 - x0 >= y1 - y0 ? on_x : on_y);
      // divide by taking half the array for each child
          (not
      // best performance with many duplicates in the
          middle)
      int half = sz(vp)/2;
      first = new Node({vp.begin(), vp.begin() + half});
      second = new Node({vp.begin() + half, vp.end()});
    }
  }
};

struct KDTree {
  Node* root;
  KDTree(const vector<P>& vp) : root(new Node({all(vp)})
      ) {}

  pair<T, P> search(Node *node, const P& p) {
    if (!node->first) {
      // uncomment if we should not find the point
          itself:
      // if (p == node->pt) return {INF, P()};
      return make_pair((p - node->pt).dist2(), node->pt)
          ;
    }

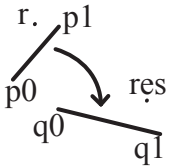
    Node *f = node->first, *s = node->second;
    T bfirst = f->distance(p), bsec = s->distance(p);
    if (bfirst > bsec) swap(bsec, bfirst), swap(f, s);

    // search closest side first, other side if needed
    auto best = search(f, p);
    if (bsec < best.first)
      best = min(best, search(s, p));
    return best;
  }

  // find nearest point to a point, and its squared
      distance
  // (requires an arbitrary operator< for Point)
  pair<T, P> nearest(const P& p) {
    return search(root, p);
  }
};
```

linearTransformation.h

Description: Apply the linear transformation (translation, rotation and scaling) which takes line p0-p1 to line q0-q1 to point r.



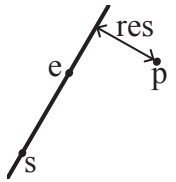
Status: not tested

"Point.h" 03a306, 6 lines

```
typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
  P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq)
      );
  return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.
      dist2();
}
```

lineDistance.h

Description: Returns the signed distance between point p and the line containing points a and b. Positive value on left side and negative on right as seen from a towards b. a==b gives nan. P is supposed to be Point<T> or Point3D<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long. Using Point3D will always give a non-negative distance. For Point3D, call .dist on the result of the cross product.



Status: tested

"Point.h" f6bf6b, 4 lines

```
template<class P>
double lineDist(const P& a, const P& b, const P& p) {
  return (double)(b-a).cross(p-a)/(b-a).dist();
}
```

LineHullIntersection.h

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon: • $(-1,-1)$ if no collision, • $(i,-1)$ if touching the corner i, • (i,i) if along side $(i,i+1)$, • (i,j) if crossing sides $(i,i+1)$ and $(j,j+1)$. In the last case, if a corner i is crossed, this is treated as happening on side $(i,i+1)$. The points are returned in the same order as the line hits the polygon. extrVertex returns the point of a hull with the max projection onto a line.
Time: $\mathcal{O}(\log n)$
Status: stress-tested

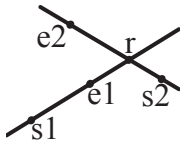
"Point.h" 7cf45b, 39 lines

```
#define cmp(i,j) sgn(dir.perp().cross(poly[(i)%n]-poly[(
    j)%n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n)
    < 0
template <class P> int extrVertex(vector<P>& poly, P dir
    ) {
  int n = sz(poly), lo = 0, hi = n;
  if (extr(0)) return 0;
  while (lo + 1 < hi) {
    int m = (lo + hi) / 2;
    if (extr(m)) return m;
    int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
    (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo
        ) = m;
  }
  return lo;
}

#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
  int endA = extrVertex(poly, (a - b).perp());
  int endB = extrVertex(poly, (b - a).perp());
  if (cmpL(endA) < 0 || cmpL(endB) > 0)
    return {-1, -1};
  array<int, 2> res;
  rep(i,0,2) {
    int lo = endB, hi = endA, n = sz(poly);
    while ((lo + 1) % n != hi) {
      int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
      (cmpL(m) == cmpL(endB) ? lo : hi) = m;
    }
    res[i] = (lo + !cmpL(hi)) % n;
    swap(endA, endB);
  }
  if (res[0] == res[1]) return {res[0], -1};
  if (!cmpL(res[0]) && !cmpL(res[1]))
    switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly))
        {
      case 0: return {res[0], res[0]};
      case 2: return {res[1], res[1]};
    }
  return res;
}
```

lineIntersection.h

Description: If a unique intersection point of the lines going through s1,e1 and s2,e2 exists {1, point} is returned. If no intersection point exists {0, (0,0)} is returned and if infinitely many exists {-1, (0,0)} is returned. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or ll.



Usage: auto res = lineInter(s1,e1,s2,e2);
if (res.first == 1)
cout << "intersection point at " << res.second << endl;
Status: stress-tested, and tested through half-plane tests

"Point.h" a01f81, 8 lines

```
template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
  auto d = (e1 - s1).cross(e2 - s2);
  if (d == 0) // if parallel
    return {-(s1.cross(e1, s2) == 0), P(0, 0)};
  auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
  return {1, (s1 * p + e1 * q) / d};
}
```

LineProjectionReflection.h

Description: Projects point p onto line ab. Set refl=true to get reflection of point p across line ab instead. The wrong point will be returned if P is an integer point and the desired point doesn't have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow.
Status: stress-tested

"Point.h" b5562d, 5 lines

```
template<class P>
P lineProj(P a, P b, P p, bool refl=false) {
  P v = b - a;
  return p - v.perp()*(1+refl)*v.cross(p-a)/v.dist2();
}
```

ManhattanMST.h

Description: Given N points, returns up to 4*N edges, which are guaranteed to contain a minimum spanning tree for the graph with edge weights w(p, q) = —p.x - q.x— + —p.y - q.y—. Edges are in the form (distance, src, dst). Use a standard MST algorithm on the result to find the final MST.
Time: $\mathcal{O}(N \log N)$
Status: Stress-tested

"Point.h" df6f59, 23 lines

```
typedef Point<int> P;
vector<array<int, 3>> manhattanMST(vector<P> ps) {
  vi id(sz(ps));
  iota(all(id), 0);
  vector<array<int, 3>> edges;
  rep(k,0,4) {
    sort(all(id), [&](int i, int j) {
       return (ps[i]-ps[j]).x < (ps[j]-ps[i]).y;});
    map<int, int> sweep;
    for (int i : id) {
      for (auto it = sweep.lower_bound(-ps[i].y);
            it != sweep.end(); sweep.erase(it++)) {
        int j = it->second;
        P d = ps[i] - ps[j];
        if (d.y > d.x) break;
        edges.push_back({d.y + d.x, i, j});
      }
      sweep[-ps[i].y] = i;
    }
    for (P& p : ps) if (k & 1) p.x = -p.x; else swap(p.x
        , p.y);
  }
  return edges;
}
```

MinimumEnclosingCircle.h

Description: Computes the minimum circle that encloses a set of points.
Time: expected $\mathcal{O}(n)$
Status: stress-tested

"circumcircle.h" 09dd0a, 17 lines

```
pair<P, double> mec(vector<P> ps) {
  shuffle(all(ps), mt19937(time(0)));
  P o = ps[0];
  double r = 0, EPS = 1 + 1e-8;
```

```
  rep(i,0,sz(ps)) if ((o - ps[i]).dist() > r * EPS) {
    o = ps[i], r = 0;
    rep(j,0,i) if ((o - ps[j]).dist() > r * EPS) {
      o = (ps[i] + ps[j]) / 2;
      r = (o - ps[i]).dist();
      rep(k,0,j) if ((o - ps[k]).dist() > r * EPS) {
        o = ccCenter(ps[i], ps[j], ps[k]);
        r = (o - ps[i]).dist();
      }
    }
  }
  return {o, r};
}
```

OnSegment.h

Description: Returns true iff p lies on the line segment from s to e. Use `(segDist(s,e,p)<=epsilon)` instead when using Point<double>.

"Point.h" c597e8, 3 lines

```
template<class P> bool onSegment(P s, P e, P p) {
  return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}
```

Point.h

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

Status: Works fine, used a lot

47ec0a, 28 lines

```
template <class T> int sgn(T x) { return (x > 0) - (x <
    0); }
template<class T>
struct Point {
  typedef Point P;
  T x, y;
  explicit Point(T x=0, T y=0) : x(x), y(y) {}
  bool operator<(P p) const { return tie(x,y) < tie(p.x,
      p.y); }
  bool operator==(P p) const { return tie(x,y)==tie(p.x,
      p.y); }
  P operator+(P p) const { return P(x+p.x, y+p.y); }
  P operator-(P p) const { return P(x-p.x, y-p.y); }
  P operator*(T d) const { return P(x*d, y*d); }
  P operator/(T d) const { return P(x/d, y/d); }
  T dot(P p) const { return x*p.x + y*p.y; }
  T cross(P p) const { return x*p.y - y*p.x; }
  T cross(P a, P b) const { return (a-*this).cross(b-*
      this); }
  T dist2() const { return x*x + y*y; }
  double dist() const { return sqrt((double)dist2()); }
  // angle to x-axis in interval [-pi, pi]
  double angle() const { return atan2(y, x); }
  P unit() const { return *this/dist(); } // makes dist
      ()=1
  P perp() const { return P(-y, x); } // rotates +90
      degrees
  P normal() const { return perp().unit(); }
  // returns point rotated 'a' radians ccw around the
      origin
  P rotate(double a) const {
    return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
  friend ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.x << "," << p.y << ")"; }
};
```

Point3D.h

Description: Class to handle points in 3D space. T can be e.g. double or long long.

Status: tested, except for phi and theta

8058ae, 32 lines

```
template<class T> struct Point3D {
  typedef Point3D P;
  typedef const P& R;
  T x, y, z;
  explicit Point3D(T x=0, T y=0, T z=0) : x(x), y(y), z(
      z) {}
  bool operator<(R p) const {
    return tie(x, y, z) < tie(p.x, p.y, p.z); }
  bool operator==(R p) const {
    return tie(x, y, z) == tie(p.x, p.y, p.z); }
  P operator+(R p) const { return P(x+p.x, y+p.y, z+p.z)
      ; }
  P operator-(R p) const { return P(x-p.x, y-p.y, z-p.z)
      ; }
  P operator*(T d) const { return P(x*d, y*d, z*d); }
  P operator/(T d) const { return P(x/d, y/d, z/d); }
  T dot(R p) const { return x*p.x + y*p.y + z*p.z; }
  P cross(R p) const {
    return P(y*p.z - z*p.y, z*p.x - x*p.z, x*p.y - y*p.x
        );
  }
  T dist2() const { return x*x + y*y + z*z; }
  double dist() const { return sqrt((double)dist2()); }
  //Azimuthal angle (longitude) to x-axis in interval [-
      pi, pi]
  double phi() const { return atan2(y, x); }
  //Zenith angle (latitude) to the z-axis in interval
      [0, pi]
  double theta() const { return atan2(sqrt(x*x+y*y),z);
      }
  P unit() const { return *this/(T)dist(); } //makes
      dist()=1
  //returns unit vector normal to *this and p
  P normal(P p) const { return cross(p).unit(); }
  //returns point rotated 'angle' radians ccw around
      axis
  P rotate(double angle, P axis) const {
    double s = sin(angle), c = cos(angle); P u = axis.
        unit();
    return u*dot(u)*(1-c) + (*this)*c - cross(u)*s;
  }
};
```

PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.

Time: $\mathcal{O}(\log N)$

Status: stress-tested

"Point.h", "sideOf.h", "OnSegment.h" 71446b, 14 lines

```
typedef Point<ll> P;

bool inHull(const vector<P>& l, P p, bool strict = true)
    {
  int a = 1, b = sz(l) - 1, r = !strict;
  if (sz(l) < 3) return r && onSegment(l[0], l.back(), p
      );
  if (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);
  if (sideOf(l[0], l[a], p) >= r || sideOf(l[0], l[b], p
      )<= -r)
    return false;
  while (abs(a - b) > 1) {
    int c = (a + b) / 2;
    (sideOf(l[0], l[c], p) > 0 ? b : a) = c;
  }
  return sgn(l[a].cross(l[b], p)) < r;
}
```

PolygonArea.h

Description: Returns twice the signed area of a polygon. Clockwise enumeration gives negative area. Watch out for overflow if using int as T!

Status: Stress-tested and tested on kattis:polygonarea

"Point.h" f12300, 6 lines

```
template<class T>
T polygonArea2(vector<Point<T>>& v) {
  T a = v.back().cross(v[0]);
  rep(i,0,sz(v)-1) a += v[i].cross(v[i+1]);
  return a;
}
```

PolygonCenter.h

Description: Returns the center of mass for a polygon.

Time: $\mathcal{O}(n)$

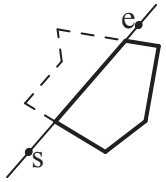
Status: Tested

"Point.h" 9706dc, 9 lines

```
typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
  P res(0, 0); double A = 0;
  for (int i = 0, j = sz(v) - 1; i < sz(v); j = i++) {
    res = res + (v[i] + v[j]) * v[j].cross(v[i]);
    A += v[j].cross(v[i]);
  }
  return res / A / 3;
}
```

PolygonCut.h

Description: Returns a vector with the vertices of a polygon with everything to the left of the line going from s to e cut away.



Usage: `vector<P> p = ...;`
`p = polygonCut(p, P(0,0), P(1,0));`
Status: tested but not extensively

"Point.h" d07181, 13 lines

```
typedef Point<double> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
  vector<P> res;
  rep(i,0,sz(poly)) {
    P cur = poly[i], prev = i ? poly[i-1] : poly.back();
    auto a = s.cross(e, cur), b = s.cross(e, prev);
    if ((a < 0) != (b < 0))
      res.push_back(cur + (prev - cur) * (a / (a - b)));
    if (a < 0)
      res.push_back(cur);
  }
  return res;
}
```

PolygonUnion.h

Description: Calculates the area of the union of n polygons (not necessarily convex). The points within each polygon must be given in CCW order. (Epsilon checks may optionally be added to sideOf/sgn, but shouldn't be needed.)
Time: $\mathcal{O}\left(N^2\right)$, where N is the total number of points
Status: stress-tested, Submitted on ECNA 2017 Problem A

"Point.h", "sideOf.h" 3931c6, 33 lines

```
typedef Point<double> P;
double rat(P a, P b) { return sgn(b.x) ? a.x/b.x : a.y/b
    .y; }
double polyUnion(vector<vector<P>>& poly) {
  double ret = 0;
  rep(i,0,sz(poly)) rep(v,0,sz(poly[i])) {
    P A = poly[i][v], B = poly[i][(v + 1) % sz(poly[i])
        ];
    vector<pair<double, int>> segs = {{0, 0}, {1, 0}};
    rep(j,0,sz(poly)) if (i != j) {
      rep(u,0,sz(poly[j])) {
        P C = poly[j][u], D = poly[j][(u + 1) % sz(poly[
            j])];
        int sc = sideOf(A, B, C), sd = sideOf(A, B, D);
        if (sc != sd) {
          double sa = C.cross(D, A), sb = C.cross(D, B);
          if (min(sc, sd) < 0)
            segs.emplace_back(sa / (sa - sb), sgn(sc -
                sd));
        } else if (!sc && !sd && j<i && sgn((B-A).dot(D-
            C))>0){
          segs.emplace_back(rat(C - A, B - A), 1);
          segs.emplace_back(rat(D - A, B - A), -1);
        }
      }
    }
    sort(all(segs));
    for (auto& s : segs) s.first = min(max(s.first, 0.0)
        , 1.0);
    double sum = 0;
    int cnt = segs[0].second;
    rep(j,1,sz(segs)) {
      if (!cnt) sum += segs[j].first - segs[j - 1].first
          ;
      cnt += segs[j].second;
    }
    ret += A.cross(B) * sum;
  }
  return ret / 2;
}
```

PolyhedronVolume.h

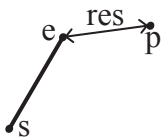
Description: Magic formula for the volume of a polyhedron. Faces should point outwards.
Status: tested

3058c3, 6 lines

```
template<class V, class L>
double signedPolyVolume(const V& p, const L& trilist) {
  double v = 0;
  for (auto i : trilist) v += p[i.a].cross(p[i.b]).dot(p
      [i.c]);
  return v / 6;
}
```

SegmentDistance.h

Description: Returns the shortest distance between point p and the line segment from point s to e.



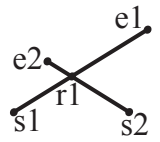
Usage: `Point<double> a, b(2,2), p(1,1);`
`bool onSegment = segDist(a,b,p) < 1e-10;`
Status: tested

"Point.h" 5c88f4, 6 lines

```
typedef Point<double> P;
double segDist(P& s, P& e, P& p) {
  if (s==e) return (p-s).dist();
  auto d = (e-s).dist2(), t = min(d,max(.0,(p-s).dot(e-s
      )));
  return ((p-s)*d-(e-s)*t).dist()/d;
}
```

SegmentIntersection.h

Description: If a unique intersection point between the line segments going from s1 to e1 and from s2 to e2 exists then it is returned. If no intersection point exists an empty vector is returned. If infinitely many exist a vector with 2 elements is returned, containing the endpoints of the common line segment. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or long long.



Usage: `vector<P> inter = segInter(s1,e1,s2,e2);`
`if (sz(inter)==1)`
`cout << "segments intersect at " << inter[0] << endl;`
Status: stress-tested, tested on kattis:intersection

"Point.h", "OnSegment.h" 9d57f2, 13 lines

```
template<class P> vector<P> segInter(P a, P b, P c, P d)
    {
  auto oa = c.cross(d, a), ob = c.cross(d, b),
    oc = a.cross(b, c), od = a.cross(b, d);
  // Checks if intersection is single non-endpoint point
      .
  if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
    return {(a * ob - b * oa) / (ob - oa)};
  set<P> s;
  if (onSegment(c, d, a)) s.insert(a);
  if (onSegment(c, d, b)) s.insert(b);
  if (onSegment(a, b, c)) s.insert(c);
  if (onSegment(a, b, d)) s.insert(d);
  return {all(s)};
}
```

sideOf.h

Description: Returns where p is as seen from s towards e. 1/0/-1 $\Leftrightarrow$ left/on line/right. If the optional argument *eps* is given 0 is returned if p is within distance *eps* from the line. P is supposed to be Point<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.
Usage: `bool left = sideOf(p1,p2,q)==1;`
Status: tested

"Point.h" 3af81c, 9 lines

```
template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }

template<class P>
int sideOf(const P& s, const P& e, const P& p, double
    eps) {
  auto a = (e-s).cross(p-s);
  double l = (e-s).dist()*eps;
  return (a > l) - (a < -l);
```

```
}
```

sphericalDistance.h

Description: Returns the shortest distance on the sphere with radius radius between the points with azimuthal angles (longitude) f1 (ϕ_1) and f2 (ϕ_2) from x axis and zenith angles (latitude) t1 (θ_1) and t2 (θ_2) from z axis (0 = north pole). All angles measured in radians. The algorithm starts by converting the spherical coordinates to cartesian coordinates so if that is what you have you can use only the two last rows. dx*radius is then the difference between the two points in the x direction and d*radius is the total distance between the points.
Status: tested on kattis:airlinehub

611f07, 8 lines

```
double sphericalDistance(double f1, double t1,
    double f2, double t2, double radius) {
  double dx = sin(t2)*cos(f2) - sin(t1)*cos(f1);
  double dy = sin(t2)*sin(f2) - sin(t1)*sin(f1);
  double dz = cos(t2) - cos(t1);
  double d = sqrt(dx*dx + dy*dy + dz*dz);
  return radius*2*asin(d/2);
}
```

Strings (9)

AhoCorasick.h

Description: Aho-corasick.
Usage: `Call init() before any operations.`
Time: $\mathcal{O}(len(s))$ for addition, $\mathcal{O}(total_len)$ for build(), $\mathcal{O}(len(x))$ or $\mathcal{O}(numofoccurences)$ for query()
Status: Tested on some problems

9db8f2, 68 lines

```
struct AhoCorasick {
  static const int SIZE = 26;
  static const char START_CHAR = 'a';

  struct Node {
    int fail, link;
    int child[SIZE], nxt[SIZE];
    int output;

    Node() {
      fail = 0, link = -1, output = 0;
      memset(child, -1, sizeof child);
      memset(nxt, -1, sizeof nxt);
    }
  };

  vector<Node> nodes;

  void init() {
    nodes.clear(), nodes.emplace_back(Node());
  }

  void add(string &s, int id) {
    int v = 0;
    for (char ch : s) {
      int c = ch - START_CHAR;
      if (nodes[v].child[c] == -1) {
        nodes[v].child[c] = nodes.size();
        nodes.emplace_back(Node());
      }
      v = nodes[v].child[c];
    }
    ++nodes[v].output;
  }

  void build() {
    nodes[0].fail = nodes[0].link = 0;
    for (int i = 0; i < SIZE; ++i)
      nodes[0].nxt[i] = (nodes[0].child[i] == -1 ? 0 :
          nodes[0].child[i]);

    queue<int> q; q.emplace(0);
    while (!q.empty()) {
      int u = q.front(); q.pop();

      for (int c = 0; c < SIZE; ++c) if (nodes[u].child[
          c] != -1) {
        int v = nodes[u].child[c];
        nodes[v].fail = (u == 0 ? 0 : nodes[nodes[u].
            fail].nxt[c]);
        for (int d = 0; d < SIZE; ++d)
          nodes[v].nxt[d] = (nodes[v].child[d] == -1 ?
              nodes[nodes[v].fail].nxt[d] : nodes[v].
              child[d]);
        if (nodes[nodes[v].fail].output != 0) nodes[v].
            link = nodes[v].fail;
        else nodes[v].link = nodes[nodes[v].fail].link;
        q.emplace(v);
      }
    }
  }

  int query(string &x) {
    int ans = 0, p = 0;
    for (auto ch : x) {
      int c = ch - START_CHAR, nxt = nodes[p].nxt[c];
      for (int u = nxt; u != 0; u = nodes[u].link) {
        ans += nodes[u].output;
      }
      p = nxt;
    }
    return ans;
  }
} aho;
```

SuffixArray.h

Description: Suffix array using prefix doubling to sort with Kasai's algorithm for LCP
Time: $\mathcal{O}\left(N \log^2 N\right)$ for construcing suffix array and $\mathcal{O}(n)$ for construcing LCP array
Status: Tested on judge.yosupo.jp

981801, 54 lines

```
struct SuffixArray {
  string str;
  vector<int> sa, lcp, order, rank;
  vector<ll> weight;
  int n;

  SuffixArray(string _str = "") {
    str = _str, n = str.size();
    sa.assign(n, 0), lcp.assign(n, 0), order.assign(n,
        0);
    rank.assign(n, 0), weight.assign(n, 0);
    getSA(), getLCP();
  }

  void getSA() {
    for (int i = 0; i < n; ++i) {
      sa[i] = i;
      weight[i] = (unsigned char)str[i];
    }

    auto cmp = [&](int x, int y) {
      return weight[x] < weight[y];
    };

    sort (sa.begin(), sa.end(), cmp);

    for (int len = 1; len <= n; len <<= 1) {
      int cnt = 0;
      for (int i = 0; i < n; ++i) {
        if (i > 0 && weight[sa[i]] == weight[sa[i - 1]])
            order[sa[i]] = order[sa[i - 1]];
        else order[sa[i]] = ++cnt;
      }
      if (cnt == n) break;
      for (int i = 0; i < n; ++i) {
        weight[i] = 1ll * order[i] * (n + 1) + (i + len
            < n ? order[i + len] : 0);
      }
      for (int i = 0, j = 0; i < n; i = j) {
        while (j < n && order[sa[j]] == order[sa[i]]) ++
            j;
        sort (sa.begin() + i, sa.begin() + j, cmp);
      }
    }
  }

  void getLCP() {
    for (int i = 0; i < n; ++i) rank[sa[i]] = i;
    for (int i = 0, k = 0; i < n; lcp[rank[i++]] = k) {
      if (rank[i] > 0) {
        int j = sa[rank[i] - 1];
        if (k > 0) --k;
        while (i + k < n && j + k < n && str[i + k] ==
            str[j + k]) ++k;
      }
      else k = 0;
```

```
    }
  }
} T;
```

KMP.h

Description: pi[x] computes the length of the longest prefix of s that ends at x, other than s[0...x] itself (abacaba -> 0010123). Can be used to find all occurrences of a string.
Time: $\mathcal{O}(n)$
Status: Tested on kattis:stringmatching

d4375c, 16 lines

```
vi pi(const string& s) {
  vi p(sz(s));
  rep(i,1,sz(s)) {
    int g = p[i-1];
    while (g && s[i] != s[g]) g = p[g-1];
    p[i] = g + (s[i] == s[g]);
  }
  return p;
}

vi match(const string& s, const string& pat) {
  vi p = pi(pat + '\0' + s), res;
  rep(i,sz(p)-sz(s),sz(p))
    if (p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
  return res;
}
```

Manacher.h

Description: For each position in a string, computes p[0][i] = half length of longest even palindrome around pos i, p[1][i] = longest odd (half rounded down).
Time: $\mathcal{O}(N)$
Status: Stress-tested

e7ad79, 13 lines

```
array<vi, 2> manacher(const string& s) {
  int n = sz(s);
  array<vi,2> p = {vi(n+1), vi(n)};
  rep(z,0,2) for (int i=0,l=0,r=0; i < n; i++) {
    int t = r-i+!z;
    if (i<r) p[z][i] = min(t, p[z][l+t]);
    int L = i-p[z][i], R = i+p[z][i]-!z;
    while (L>=1 && R+1<n && s[L-1] == s[R+1])
      p[z][i]++, L--, R++;
    if (R>r) l=L, r=R;
  }
  return p;
}
```

MinRotation.h

Description: Finds the lexicographically smallest rotation of a string.
Usage: `rotate(v.begin(), v.begin()+minRotation(v), v.end());`
Time: $\mathcal{O}(N)$
Status: Stress-tested

d07a42, 8 lines

```
int minRotation(string s) {
  int a=0, N=sz(s); s += s;
  rep(b,0,N) rep(k,0,N) {
    if (a+k == b || s[a+k] < s[b+k]) {b += max(0, k-1);
        break;}
    if (s[a+k] > s[b+k]) { a = b; break; }
  }
  return a;
}
```

PalindromicTree.h

Description: Builds palindromic tree for a string.
Status: stress-tested

f8784d, 22 lines

```
namespace PalindromicTree {
const int N = 100000 + 5, sigma = 26;
int s[N], len[N], link[N], to[N][sigma], n, last, sz;
void init() {
  fill(len, len + sz, 0); fill(link, link + sz, 0);
  for (int i = 0; i < sz; ++i) fill(to[i], to[i] + sigma
      , 0);
  n = last = 0; s[n++] = -1; link[0] = 1; len[1] = -1;
      sz = 2;
}
int get_link(int v) {
  while (s[n - len[v] - 2] != s[n - 1]) v = link[v];
  return v;
}
void add_letter(int c) {
  s[n++] = c; last = get_link(last);
  if (!to[last][c]) {
    len[sz] = len[last] + 2;
    link[sz] = to[get_link(link[last])][c];
    to[last][c] = sz++;
  }
  last = to[last][c];
}
}
```

SuffixAutomaton.h

Description: Suffix Automaton, supporting multiple strings. The beginning iterator is 0. Make calls to itmove to progress iterator.
Time: $\mathcal{O}(n)$
Status: stress-tested

be95c5, 29 lines

```
struct SuffixAutomaton{
  struct Node{
    int len, link = -1, nxt[26];
    Node(int len): len(len) {fill(nxt, nxt + 26, -1);}
    Node(int len, int link, int _nxt[26]):
    len(len), link(link) {memcpy(nxt, _nxt, sizeof nxt);
        }
  };
  vector<Node> g = {0};
  int ptrans(int p, char c, int nxt) {
    for (int q = g[p].nxt[c]; p != -1 && g[p].nxt[c] ==
        q;
      p = g[p].link) g[p].nxt[c] = nxt;
    return p;
  }
  int fbuild(int p, char c) {
    if (p == -1) return 0;
    int q = g[p].nxt[c];
    if (g[p].len + 1 == g[q].len) return q;
    int qc = g.size();
    g.emplace_back(g[p].len+1, g[q].link, g[q].nxt);
    ptrans(p, c, qc); return g[q].link = qc;
  }
  int addc(int p, char c) { // Progress iterator; c is
      in [0, 26).
    if (g[p].nxt[c] != -1) return fbuild(p, c);
    int nc = g.size();
    g.emplace_back(g[p].len+1);
    int r = fbuild(ptrans(p, c, nc), c);
    g[nc].link = r; return nc;
  }
};
```

SuffixTree.h

Description: Ukkonen's algorithm for online suffix tree construction. Each node contains indices [l, r) into the string, and a list of child nodes. Suffixes are given by traversals of this tree, joining [l, r) substrings. The root is 0 (has l = -1, r = 0), non-existent children are -1. To get a complete tree, append a dummy symbol – otherwise it may contain an incomplete path (still useful for substring matching, though).
Time: $\mathcal{O}(26N)$
Status: stress-tested a bit

aae0b8, 50 lines

```
struct SuffixTree {
  enum { N = 200010, ALPHA = 26 }; // N ~ 2*maxlen+10
  int toi(char c) { return c - 'a'; }
  string a; // v = cur node, q = cur position
  int t[N][ALPHA],l[N],r[N],p[N],s[N],v=0,q=0,m=2;

  void ukkadd(int i, int c) { suff:
    if (r[v]<=q) {
      if (t[v][c]==-1) { t[v][c]=m;  l[m]=i;
        p[m++]=v; v=s[v]; q=r[v];  goto suff; }
      v=t[v][c]; q=l[v];
    }
    if (q==-1 || c==toi(a[q])) q++; else {
      l[m+1]=i;  p[m+1]=m;  l[m]=l[v];  r[m]=q;
      p[m]=p[v];  t[m][c]=m+1;  t[m][toi(a[q])]=v;
      l[v]=q;  p[v]=m;  t[p[m]][toi(a[l[m]])]=m;
      v=s[p[m]];  q=l[m];
      while (q<r[m]) { v=t[v][toi(a[q])];  q+=r[v]-l[v];
          }
      if (q==r[m])  s[m]=v;  else s[m]=m+2;
      q=r[v]-(q-r[m]);  m+=2;  goto suff;
    }
  }

  SuffixTree(string a) : a(a) {
    fill(r,r+N,sz(a));
    memset(s, 0, sizeof s);
```

```
    memset(t, -1, sizeof t);
    fill(t[1],t[1]+ALPHA,0);
    s[0] = 1; l[0] = l[1] = -1; r[0] = r[1] = p[0] = p
        [1] = 0;
    rep(i,0,sz(a)) ukkadd(i, toi(a[i]));
  }

  // example: find longest common substring (uses ALPHA
      = 28)
  pii best;
  int lcs(int node, int i1, int i2, int olen) {
    if (l[node] <= i1 && i1 < r[node]) return 1;
    if (l[node] <= i2 && i2 < r[node]) return 2;
    int mask = 0, len = node ? olen + (r[node] - l[node
        ]) : 0;
    rep(c,0,ALPHA) if (t[node][c] != -1)
      mask |= lcs(t[node][c], i1, i2, len);
    if (mask == 3)
      best = max(best, {len, r[node] - len});
    return mask;
  }
  static pii LCS(string s, string t) {
    SuffixTree st(s + (char)('z' + 1) + t + (char)('z' +
        2));
    st.lcs(0, sz(s), sz(s) + 1 + sz(t), 0);
    return st.best;
  }
};
```

Zfunc.h

Description: z[i] computes the length of the longest common prefix of s[i:] and s, except z[0] = 0. (abacaba -> 0010301)
Time: $\mathcal{O}(n)$
Status: stress-tested

ee09e2, 12 lines

```
vi Z(const string& S) {
  vi z(sz(S));
  int l = -1, r = -1;
  rep(i,1,sz(S)) {
    z[i] = i >= r ? 0 : min(r - i, z[i - l]);
    while (i + z[i] < sz(S) && S[i + z[i]] == S[z[i]])
      z[i]++;
    if (i + z[i] > r)
      l = i, r = i + z[i];
  }
  return z;
}
```

Various (10)

FastMod.h

Description: Compute $a\%b$ about 5 times faster than usual, where b is constant but not known at compile time. Returns a value congruent to a (mod b) in the range $[0, 2b)$.
Status: proven correct, stress-tested

62d9a9, 7 lines

```
struct FastMod {
  uint64_t b, m;
  FastMod(uint64_t b) : b(b), m(-1ULL / b) {}
  uint64_t reduce(uint64_t a) { // a % b + (0 or b)
    return a - (uint64_t)((__uint128_t(m) * a) >> 64) *
        b;
  }
};
```

ConstantIntervals.h

Description: Split a monotone function on [from, to) into a minimal set of half-open intervals on which it has the same value. Runs a callback g for each such interval.
Usage: `constantIntervals(0, sz(v), [&](int x){return v[x];}, [&](int lo, int hi, T val){...});`
Time: $\mathcal{O}\left(k \log \frac{n}{k}\right)$
Status: tested

753a4c, 19 lines

```
template<class F, class G, class T>
void rec(int from, int to, F& f, G& g, int& i, T& p, T q
    ) {
  if (p == q) return;
  if (from == to) {
    g(i, to, p);
    i = to; p = q;
  } else {
    int mid = (from + to) >> 1;
    rec(from, mid, f, g, i, p, f(mid));
    rec(mid+1, to, f, g, i, p, q);
  }
}
template<class F, class G>
void constantIntervals(int from, int to, F f, G g) {
  if (to <= from) return;
  int i = from; auto p = f(i), q = f(to-1);
  rec(from, to-1, f, g, i, p, q);
  g(i, to, q);
}
```

DivideAndConquerDP.h

Description: Given $a[i] = \min_{lo(i) \le k < hi(i)}(f(i,k))$ where the (minimal) optimal k increases with i, computes $a[i]$ for $i = L..R-1$.
Time: $\mathcal{O}\left((N + (hi - lo)) \log N\right)$
Status: tested on http://codeforces.com/contest/321/problem/E

d38d2b, 18 lines

```
struct DP { // Modify at will:
  int lo(int ind) { return 0; }
  int hi(int ind) { return ind; }
  ll f(int ind, int k) { return dp[ind][k]; }
  void store(int ind, int k, ll v) { res[ind] = pii(k, v
      ); }

  void rec(int L, int R, int LO, int HI) {
    if (L >= R) return;
    int mid = (L + R) >> 1;
    pair<ll, int> best(LLONG_MAX, LO);
    rep(k, max(LO,lo(mid)), min(HI,hi(mid)))
      best = min(best, make_pair(f(mid, k), k));
    store(mid, best.second, best.first);
    rec(L, mid, LO, best.second+1);
    rec(mid+1, R, best.second, HI);
  }
  void solve(int L, int R) { rec(L, R, INT_MIN, INT_MAX)
      ; }
};
```

FastInput.h

Description: Read an integer from stdin. Usage requires your program to pipe in input from file.
Usage: `./a.out < input.txt`
Time: About 5x as fast as cin/scanf.
Status: tested on SPOJ INTEST, unit tested

7b3c70, 16 lines

```
inline char gc() { // like getchar()
  static char buf[1 << 16];
  static size_t bc, be;
  if (bc >= be) {
    buf[0] = 0, bc = 0;
    be = fread(buf, 1, sizeof(buf), stdin);
  }
  return buf[bc++]; // returns 0 on EOF
}
int readInt() {
  int a, c;
  while ((a = gc()) < 40);
  if (a == '-') return -readInt();
  while ((c = gc()) >= 48) a = a * 10 + c - 480;
  return a - 48;
}
```

FastKnapsack.h

Description: Given N non-negative integer weights w and a non-negative target t, computes the maximum S <= t such that S is the sum of some subset of the weights.
Time: $\mathcal{O}(N \max(w_i))$
Status: Tested on kattis:eavesdropperevasion, stress-tested

b20ccc, 16 lines

```
int knapsack(vi w, int t) {
  int a = 0, b = 0, x;
  while (b < sz(w) && a + w[b] <= t) a += w[b++];
  if (b == sz(w)) return a;
  int m = *max_element(all(w));
  vi u, v(2*m, -1);
  v[a+m-t] = b;
  rep(i,b,sz(w)) {
    u = v;
    rep(x,0,m) v[x+w[i]] = max(v[x+w[i]], u[x]);
    for (x = 2*m; --x > m;) rep(j, max(0,u[x]), v[x])
      v[x-w[j]] = max(v[x-w[j]], j);
  }
  for (a = t; v[a+m-t] < 0; a--) ;
  return a;
}
```

IntervalContainer.h

Description: Add and remove intervals from a set of disjoint intervals. Will merge the added interval with any overlapping intervals in the set when adding. Intervals are [inclusive, exclusive).
Time: $\mathcal{O}(\log N)$
Status: stress-tested

edce47, 22 lines

```
set<pii>::iterator addInterval(set<pii>& is, int L, int
    R) {
  if (L == R) return is.end();
  auto it = is.lower_bound({L, R}), before = it;
  while (it != is.end() && it->first <= R) {
    R = max(R, it->second);
    before = it = is.erase(it);
  }
  if (it != is.begin() && (--it)->second >= L) {
    L = min(L, it->first);
    R = max(R, it->second);
    is.erase(it);
  }
  return is.insert(before, {L,R});
}
void removeInterval(set<pii>& is, int L, int R) {
  if (L == R) return;
  auto it = addInterval(is, L, R);
  auto r2 = it->second;
  if (it->first == L) is.erase(it);
  else (int&)it->second = L;
  if (R != r2) is.emplace(R, r2);
}
```

IntervalCover.h

Description: Compute indices of smallest set of intervals covering another interval. Intervals should be [inclusive, exclusive). To support [inclusive, inclusive], change (A) to add `|| R.empty()`. Returns empty set on failure (or if G is empty).
Time: $\mathcal{O}(N \log N)$
Status: Tested on kattis:intervalcover

9e9d8d, 19 lines

```
template<class T>
vi cover(pair<T, T> G, vector<pair<T, T>> I) {
  vi S(sz(I)), R;
  iota(all(S), 0);
  sort(all(S), [&](int a, int b) { return I[a] < I[b]; }
      );
  T cur = G.first;
  int at = 0;
  while (cur < G.second) { // (A)
    pair<T, int> mx = make_pair(cur, -1);
    while (at < sz(I) && I[S[at]].first <= cur) {
      mx = max(mx, make_pair(I[S[at]].second, S[at]));
      at++;
    }
    if (mx.second == -1) return {};
    cur = mx.first;
    R.push_back(mx.second);
  }
  return R;
}
```

KnuthDP.h

Description: When doing DP on intervals: $a[i][j] = \min_{i<k<j}(a[i][k] + a[k][j]) + f(i,j)$, where the (minimal) optimal k increases with both i and j, one can solve intervals in increasing order of length, and search $k = p[i][j]$ for $a[i][j]$ only between $p[i][j-1]$ and $p[i+1][j]$. This is known as Knuth DP. Sufficient criteria for this are if $f(b,c) \le f(a,d)$ and $f(a,c) + f(b,d) \le f(a,d) + f(b,c)$ for all $a \le b \le c \le d$. Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.
Time: $\mathcal{O}\left(N^2\right)$

SIMD.h

Description: Cheat sheet of SSE/AVX intrinsics, for doing arithmetic on several numbers at once. Can provide a constant factor improvement of about 4, orthogonal to loop unrolling. Operations follow the pattern `"_mm(256)?_name_(si(128|256)|epi(8|16|32|64)|pd|ps)"`. Not all are described here; grep for `_mm_` in `/usr/lib/gcc/*/4.9/include/` for more. If AVX is unsupported, try 128-bit operations, "emmintrin.h" and #define `__SSE__` and `__MMX__` before including it. For aligned memory use `_mm_malloc(size, 32)` or `int buf[N] alignas(32)`, but prefer loadu/storeu.

c9ac08, 43 lines

```
#pragma GCC target ("avx2") // or sse4.1
#include "immintrin.h"

typedef __m256i mi;
#define L(x) _mm256_loadu_si256((mi*)&(x))

// High-level/specific methods:
// load(u)?_si256, store(u)?_si256, setzero_si256,
    _mm_malloc
// blendv_(epi8|ps|pd) (z?y:x), movemask_epi8 (hibits of
    bytes)
// i32gather_epi32(addr, x, 4): map addr[] over 32-b
    parts of x
// sad_epu8: sum of absolute differences of u8, outputs
    4xi64
// maddubs_epi16: dot product of unsigned i7's, outputs
    16xi15
// madd_epi16: dot product of signed i16's, outputs 8
    xi32
// extractf128_si256(, i) (256->128), cvtsi128_si32
    (128->lo32)
// permute2f128_si256(x,x,1) swaps 128-bit lanes
// shuffle_epi32(x, 3*64+2*16+1*4+0) == x for each lane
// shuffle_epi8(x, y) takes a vector instead of an imm

// Methods that work with most data types (append e.g.
    _epi32):
// set1, blend (i8?x:y), add, adds (sat.), mullo, sub,
    and/or,
// andnot, abs, min, max, sign(1,x), cmp(gt|eq), unpack(
    lo|hi)

int sumi32(mi m) { union {int v[8]; mi m;} u; u.m = m;
  int ret = 0; rep(i,0,8) ret += u.v[i]; return ret; }
mi zero() { return _mm256_setzero_si256(); }
mi one() { return _mm256_set1_epi32(-1); }
bool all_zero(mi m) { return _mm256_testz_si256(m, m); }
bool all_one(mi m) { return _mm256_testc_si256(m, one())
    ; }

ll example_filteredDotProduct(int n, short* a, short* b)
    {
  int i = 0; ll r = 0;
  mi zero = _mm256_setzero_si256(), acc = zero;
  while (i + 16 <= n) {
    mi va = L(a[i]), vb = L(b[i]); i += 16;
    va = _mm256_and_si256(_mm256_cmpgt_epi16(vb, va), va
        );
    mi vp = _mm256_madd_epi16(va, vb);
    acc = _mm256_add_epi64(_mm256_unpacklo_epi32(vp,
        zero),
      _mm256_add_epi64(acc, _mm256_unpackhi_epi32(vp,
          zero)));
  }
  union {ll v[4]; mi m;} u; u.m = acc; rep(i,0,4) r += u
      .v[i];
  for (;i<n;++i) if (a[i] < b[i]) r += a[i]*b[i]; // <-
      equiv
  return r;
}
```

TernarySearch.h

Description: Find the smallest i in $[a,b]$ that maximizes $f(i)$, assuming that $f(a) < \ldots < f(i) \ge \cdots \ge f(b)$. To reverse which of the sides allows non-strict inequalities, change the < marked with (A) to <=, and reverse the loop at (B). To minimize f, change it to >, also at (B).
Usage: `int ind = ternSearch(0,n-1,[&](int i){return a[i];});`
Time: $\mathcal{O}(\log(b-a))$
Status: tested

9155b4, 11 lines

```
template<class F>
int ternSearch(int a, int b, F f) {
  assert(a <= b);
  while (b - a >= 5) {
    int mid = (a + b) / 2;
    if (f(mid) < f(mid+1)) a = mid; // (A)
    else b = mid+1;
  }
  rep(i,a+1,b+1) if (f(a) < f(i)) a = i; // (B)
  return a;
}
```